

IAT461_FinalProject_Tawshif_Al_Mehrab

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1 SerafinoSignal

1.1 Restaurant Complaint Theme Discovery Across Restaurants in Turkey

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Course: IAT 461

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1.2 1. Project Overview

This project examines restaurant reviews from a hypothetical Turkish restaurant review website. The Director of Marketing and Outreach is the stakeholder, and instead of depending solely on unfiltered reviews and star ratings, he wants to know the recurrent causes of customer dissatisfaction. The main question is:

What review experience themes appear across restaurant reviews and which themes recur across multiple restaurants rather than being specific to one restaurant?

Due to the lack of predefined complaint theme labels in the dataset, the project employs unsupervised text analysis. Finding significant and useful complaint patterns in the review text is the aim.

AI Usage declaration - Purpose: This notebook uses Perplexity AI to assist with coding specific cells, generate tables, ideas, polishing brainstorm approaches and enhance results. All AI-proposed code is reviewed and edited by me before execution. Anything committed as final should include a note: 'AI-assisted draft; review before use.'

Brainstorm ideas outline (expand ideas during sessions) - Problem framing: Restate the goal and success criteria in 1-2 sentences. - Quick experiments: Propose 2-3 small cell experiments to test viability.

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##All Imports and Libraries

```
[1]: import re
import hdbscan
import umap.umap_ as umap
import hdbscan
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sentence_transformers import SentenceTransformer
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from pathlib import Path
from sklearn.metrics.pairwise import cosine_similarity
os.environ['OMP_NUM_THREADS'] = '5'

!pip install umap-learn
!pip install hdbscan
```

Requirement already satisfied: umap-learn in C:\Users\tawsi\miniconda3\Lib\site-packages (0.5.12)

Requirement already satisfied: numpy>=1.23 in C:\Users\tawsi\miniconda3\Lib\site-packages (from umap-learn) (2.4.4)

Requirement already satisfied: scipy>=1.3.1 in C:\Users\tawsi\miniconda3\Lib\site-packages (from umap-learn) (1.17.1)

Requirement already satisfied: scikit-learn>=1.6 in C:\Users\tawsi\miniconda3\Lib\site-packages (from umap-learn) (1.7.1)

Requirement already satisfied: numba>=0.51.2 in C:\Users\tawsi\miniconda3\Lib\site-packages (from umap-learn) (0.66.0)

Requirement already satisfied: pynndescent>=0.5 in C:\Users\tawsi\miniconda3\Lib\site-packages (from umap-learn) (0.6.0)

Requirement already satisfied: tqdm in C:\Users\tawsi\miniconda3\Lib\site-

```

packages (from umap-learn) (4.67.3)
Requirement already satisfied: llvmlite<0.49,>=0.48.0dev0 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from numba>=0.51.2->umap-learn)
(0.48.0)
Requirement already satisfied: joblib>=0.11 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from pynndescent>=0.5->umap-learn)
(1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from scikit-learn>=1.6->umap-learn)
(3.5.0)
Requirement already satisfied: colorama in C:\Users\tawsi\miniconda3\Lib\site-
packages (from tqdm->umap-learn) (0.4.6)
Requirement already satisfied: hdbscan in C:\Users\tawsi\miniconda3\Lib\site-
packages (0.8.44)
Requirement already satisfied: numpy<3,>=1.20 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from hdbscan) (2.4.4)
Requirement already satisfied: scipy>=1.0 in C:\Users\tawsi\miniconda3\Lib\site-
packages (from hdbscan) (1.17.1)
Requirement already satisfied: scikit-learn>=1.6 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from hdbscan) (1.7.1)
Requirement already satisfied: joblib>=1.0 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from hdbscan) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
C:\Users\tawsi\miniconda3\Lib\site-packages (from scikit-learn>=1.6->hdbscan)
(3.5.0)

```

```

[2]: pd.set_option('display.max_colwidth', 150)
DATA_PATH = 'reviews.csv'
df = pd.read_csv(DATA_PATH)
print('Dataset shape:', df.shape)
display(df.head())

```

Dataset shape: (1100, 6)

	business_name	author_name \
0	Haci'nin Yeri - Yigit Lokantasi	Gulsum Akar
1	Haci'nin Yeri - Yigit Lokantasi	Oguzhan Cetin
2	Haci'nin Yeri - Yigit Lokantasi	Yasin Kuyu
3	Haci'nin Yeri - Yigit Lokantasi	Orhan Kapu
4	Haci'nin Yeri - Yigit Lokantasi	Ozgur Sati

	text \
0	We went to Marmaris with my wife for a holiday. We chose this restaurant as a place for dinner based on the reviews and because we wanted juicy fo...
1	During my holiday in Marmaris we ate here to fit the food. It's really good that the food is cheap and nice. Eating as much bread as you want is a...

2 Prices are very affordable. The menu in the photo cost 108 liras. You have to
 ↳wait 10-15 minutes for food. Staff is annoying. Well it tastes good...
 3
 ↳ Turkey's cheapest artisan restaurant and its food is delicious!
 4 I don't know what you will look for in terms of price performance point;↳
 ↳taste; but yigit restaurant writes a big plus for those who come to work ...

	photo	rating \
0	dataset/taste/hacinin_yeri_gulsum_akar.png	5
1	dataset/menu/hacinin_yeri_oguzhan_cetin.png	4
2	dataset/outdoor_atmosphere/hacinin_yeri_yasin_kuyu.png	3
3	dataset/indoor_atmosphere/hacinin_yeri_orhan_kapu.png	5
4	dataset/menu/hacinin_yeri_ozgur_sati.png	3

	rating_category
0	taste
1	menu
2	outdoor_atmosphere
3	indoor_atmosphere
4	menu

```
[3]: print('Columns:')
print(df.columns.tolist())
print('\nData types:')
display(df.dtypes.to_frame('data_type'))
print('\nMissing values:')
display(df.isna().sum().to_frame('missing_values'))
print('\nUnique restaurants:')
print(df['business_name'].nunique())
print('\nExact duplicate rows:')
print(df.duplicated().sum())
print('\nDuplicate review texts:')
print(df.duplicated(subset=['text']).sum())
```

Columns:

```
['business_name', 'author_name', 'text', 'photo', 'rating', 'rating_category']
```

Data types:

	data_type
business_name	str
author_name	str
text	str
photo	str
rating	int64
rating_category	str

Missing values:

	missing_values
business_name	0
author_name	0
text	0
photo	0
rating	0
rating_category	0

Unique restaurants:
100

Exact duplicate rows:
0

Duplicate review texts:
12

```
[4]: print('Rating distribution:')
display(df['rating'].value_counts().sort_index().to_frame('reviews'))
print('\nRating category distribution:')
display(df['rating_category'].value_counts().to_frame('reviews'))
print('\nReviews per restaurant:')
display(df['business_name'].value_counts().describe())
```

Rating distribution:

	reviews
rating	
1	80
2	72
3	172
4	316
5	460

Rating category distribution:

	reviews
rating_category	
taste	330
indoor_atmosphere	288
outdoor_atmosphere	286
menu	196

Reviews per restaurant:

count	100.0
mean	11.0
std	0.0

```

min          11.0
25%          11.0
50%          11.0
75%          11.0
max          11.0
Name: count, dtype: float64

```

```

[5]: df['review_word_count'] = ( df['text'].astype(str).str.split().str.len())
      display(df['review_word_count'].describe())

```

```

count      1100.000000
mean        20.051818
std         12.977619
min          1.000000
25%         11.000000
50%         19.000000
75%         27.000000
max         179.000000
Name: review_word_count, dtype: float64

```

```

[6]: rating_groups = pd.cut(df['rating'],bins=[0, 2, 3, 5],labels=['NEGATIVE_
      ↪(1-2)', 'MIXED (3)', 'POSITIVE (4-5)'])
      df['rating_group'] = rating_groups
      display(df['rating_group'].value_counts().to_frame('reviews'))

```

	reviews
rating_group	
POSITIVE (4-5)	776
MIXED (3)	172
NEGATIVE (1-2)	152

All reviews are included in the analysis based on client clarification. Ratings will not be used to filter the corpus before clustering. Instead, ratings will be used afterward to understand whether discovered themes are associated with positive, negative, or mixed experiences.

1.4 2. Exploratory Data Analysis

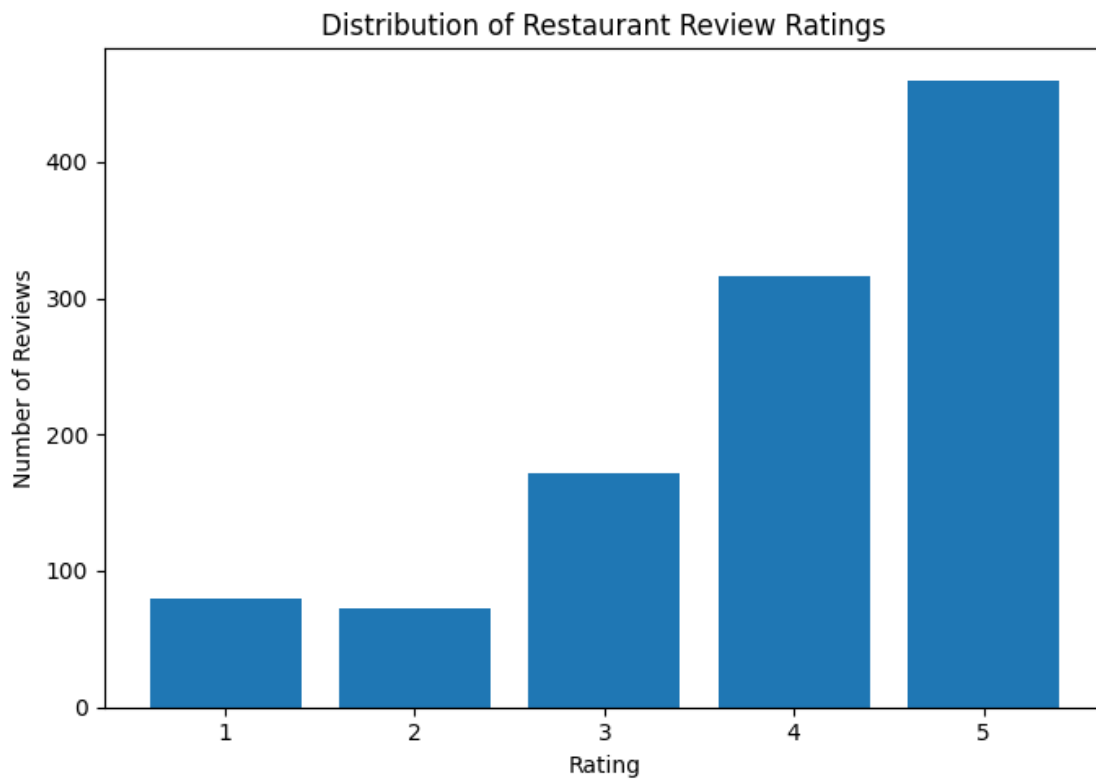
Prior to using unsupervised learning, this section looks at the overall structure of the review dataset with the aim of comprehending rating patterns, review length, restaurant level variations, and current review categories. For the purpose of theme discovery, all reviews are kept; ratings are then utilized to determine whether a theme is linked to favourable, conflicting, or unfavourable experiences.

```

[7]: rating_counts = (df['rating'].value_counts().sort_index())
      plt.figure(figsize=(7, 5))
      plt.bar(rating_counts.index.astype(str),rating_counts.values)
      plt.xlabel('Rating')
      plt.ylabel('Number of Reviews')
      plt.title('Distribution of Restaurant Review Ratings')

```

```
plt.tight_layout()
plt.show()
```



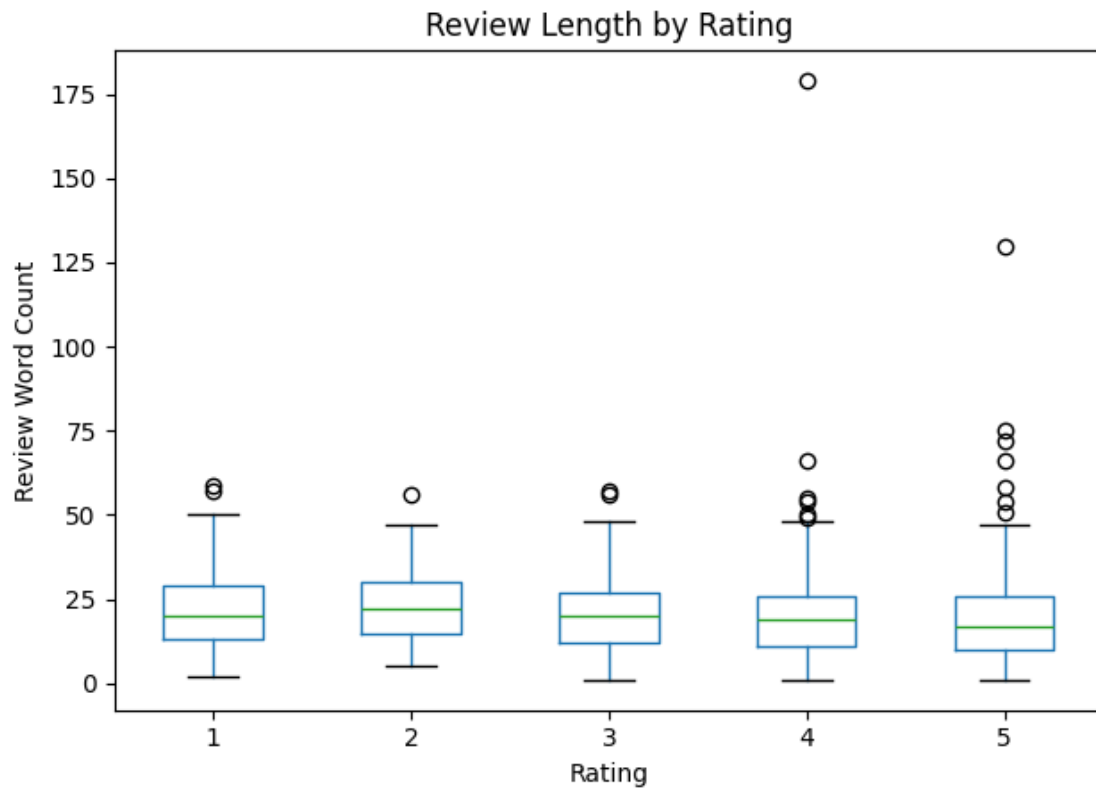
```
[8]: review_length_by_rating = (df.groupby('rating')['review_word_count'].
    ↪agg(['mean', 'median', 'count']) .round(2))
display(review_length_by_rating)
```

	mean	median	count
rating			
1	21.76	20.0	80
2	23.36	22.0	72
3	20.14	20.0	172
4	20.12	19.0	316
5	19.16	17.0	460

```
[9]: plt.figure(figsize=(8, 5))
df.boxplot(column='review_word_count',by='rating',grid=False)
plt.xlabel('Rating')
plt.ylabel('Review Word Count')
plt.title('Review Length by Rating')
plt.suptitle('')
plt.tight_layout()
```

```
plt.show()
```

<Figure size 800x500 with 0 Axes>



```
[10]: category_counts = (df['rating_category'].value_counts())
plt.figure(figsize=(8, 5))
plt.bar( category_counts.index,category_counts.values)
plt.xlabel('Rating Category')
plt.ylabel('Number of Reviews')
plt.title('Distribution of Existing Review Categories')
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```




```
[11]: category_rating_summary = (df.groupby('rating_category')['rating'].agg(['mean',
    ↳ 'median', 'count']).sort_values('mean').round(2))
display(category_rating_summary)
```

	mean	median	count
rating_category			
menu	3.69	4.0	196
taste	3.94	4.0	330
indoor_atmosphere	3.96	4.0	288
outdoor_atmosphere	3.99	4.0	286

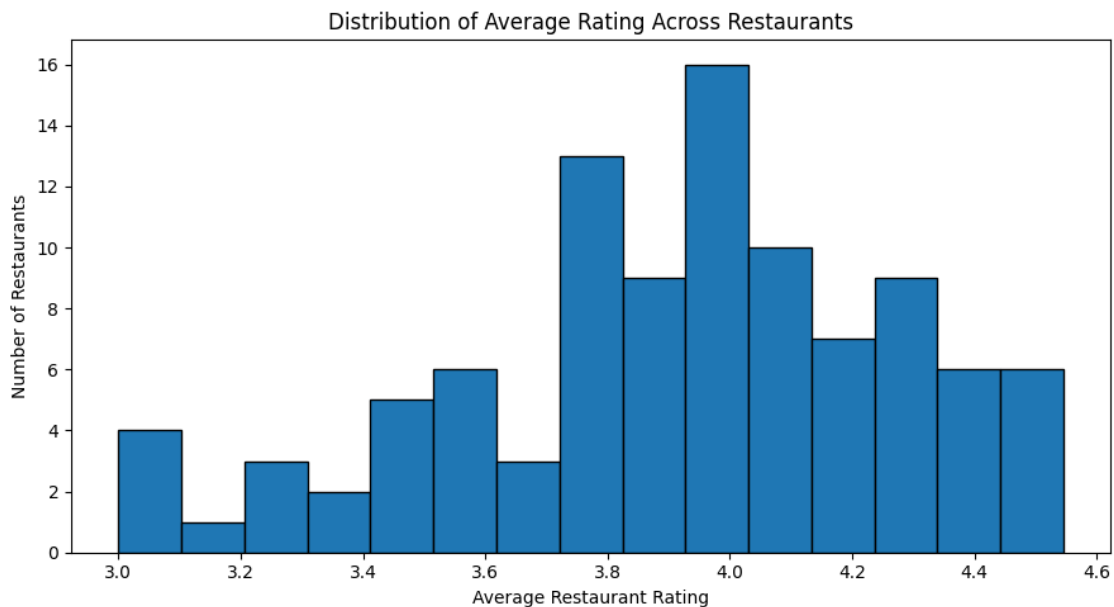
```
[12]: restaurant_summary = (df.groupby('business_name').agg(reviews=('text',
    ↳ 'count'),average_rating=('rating', 'mean'),median_rating=('rating',
    ↳ 'median')).sort_values('average_rating'))
display(restaurant_summary.head(10))
display(restaurant_summary.tail(10))
```

	reviews	average_rating	median_rating
business_name			
Medeniyetler Sofrasi	11	3.000000	3.0
Popeyes	11	3.000000	3.0
Salih Usta	11	3.000000	3.0
Baskent Aspava	11	3.090909	3.0
Mado	11	3.181818	3.0
Tavuk Dunyasi	11	3.272727	4.0
KFC	11	3.272727	4.0

Diyarbakir Ocakbasi Palace	11	3.272727	4.0
Cemo Etliemek	11	3.363636	4.0
Mcdonalds Fethiye	11	3.363636	4.0

	reviews	average_rating	median_rating
business_name			
Hasan Usta	11	4.363636	5.0
Pukka Marmaris	11	4.363636	4.0
Cici Tantuni	11	4.363636	4.0
Kumsal Lokantasi	11	4.363636	4.0
Bunbun Kas	11	4.454545	5.0
Churchill Bistro	11	4.454545	5.0
Haci'nin Yeri - Yigit Lokantasi	11	4.454545	5.0
Antakya Mutfagi	11	4.454545	5.0
Yesemek	11	4.454545	5.0
Pizzeria Remo	11	4.545455	5.0

```
[13]: plt.figure(figsize=(9, 5))
plt.hist( restaurant_summary['average_rating'],bins=15,edgecolor='black')
plt.xlabel('Average Restaurant Rating')
plt.ylabel('Number of Restaurants')
plt.title('Distribution of Average Rating Across Restaurants')
plt.tight_layout()
plt.show()
```



```
[14]: duplicate_texts = df[df.duplicated(subset=['text'],keep=False)
].sort_values('text')
```

```
print('Rows involved in duplicate review texts:', len(duplicate_texts))
display(duplicate_texts[['business_name', 'text', 'rating', 'rating_category']])
```

Rows involved in duplicate review texts: 19

	business_name	text	rating	rating_category
162	Tatbak	Delicious!	4	taste
321	Sardalye	Delicious!	5	indoor_atmosphere
962	Kadineli	Delicious!	5	outdoor_atmosphere
942	Fikret Balik Restaurant	Expensive place.	1	menu
1050	Lipa	Expensive place.	3	menu
500	Kebabci Halil Usta	Expensive.	3	outdoor_atmosphere
619	Midyeci Ahmet	Expensive.	3	menu
973	Anzelha	Expensive.	3	indoor_atmosphere
72	Gold Semaver	Great.	4	outdoor_atmosphere
228	Salacak	Great.	4	menu
424	Bilice	Great.	5	indoor_atmosphere
314	Sushico	Liked.	4	taste
544	Askin Tostevi	Liked.	4	outdoor_atmosphere
685	Duveroglu	Liked.	5	outdoor_atmosphere
263	Cici Tantuni	Not bad.	3	taste
356	Bogazkoy Restaurant	Not bad.	3	outdoor_atmosphere
296	Nazar Cafe	Recommend.	4	indoor_atmosphere
534	Kasikla Restaurant	Recommend.	5	outdoor_atmosphere
818	Tavuk Dunyasi	Recommend.	5	outdoor_atmosphere

1.5 3. Data Cleaning Text Preparation

The dataset has no missing values or completely duplicated rows, according to this exploratory analysis. A few review texts are repeated across different restaurants, but they are primarily brief and general statements.

```
[15]: model_df = df[['business_name', 'text', 'rating', 'rating_category', 'rating_group',
    ↪ 'review_word_count']].copy()
model_df = model_df.rename( columns={'text': 'text_original'})
print('Modeling dataset shape:', model_df.shape)
display(model_df.head())
```

Modeling dataset shape: (1100, 6)

	business_name \
0	Haci'nin Yeri - Yigit Lokantasi
1	Haci'nin Yeri - Yigit Lokantasi
2	Haci'nin Yeri - Yigit Lokantasi
3	Haci'nin Yeri - Yigit Lokantasi
4	Haci'nin Yeri - Yigit Lokantasi

↪

text_original \

↪

0 We went to Marmaris with my wife for a holiday. We chose this restaurant as a
 ↳ place for dinner based on the reviews and because we wanted juicy fo...
 1 During my holiday in Marmaris we ate here to fit the food. It's really good
 ↳ that the food is cheap and nice. Eating as much bread as you want is a...
 2 Prices are very affordable. The menu in the photo cost 108 liras. You have to
 ↳ wait 10-15 minutes for food. Staff is annoying. Well it tastes good...
 3
 ↳ Turkey's cheapest artisan restaurant and its food is delicious!
 4 I don't know what you will look for in terms of price performance point;
 ↳ taste; but yigit restaurant writes a big plus for those who come to work ...

	rating	rating_category	rating_group	review_word_count
0	5	taste	POSITIVE (4-5)	130
1	4	menu	POSITIVE (4-5)	179
2	3	outdoor_atmosphere	MIXED (3)	31
3	5	indoor_atmosphere	POSITIVE (4-5)	9
4	3	menu	MIXED (3)	31

```
[16]: def clean_review_text(text):
      text = str(text)
      text = re.sub(r'\s+', ' ', text)
      text = re.sub(r'\s+([.!?;:])', r'\1', text)
      return text.strip()
model_df['text_clean'] = (model_df['text_original'].apply(clean_review_text))
display(model_df[['text_original', 'text_clean']].head(10))
```

↳ text_original \

0 We went to Marmaris with my wife for a holiday. We chose this restaurant as a
 ↳ place for dinner based on the reviews and because we wanted juicy fo...
 1 During my holiday in Marmaris we ate here to fit the food. It's really good
 ↳ that the food is cheap and nice. Eating as much bread as you want is a...
 2 Prices are very affordable. The menu in the photo cost 108 liras. You have to
 ↳ wait 10-15 minutes for food. Staff is annoying. Well it tastes good...
 3
 ↳ Turkey's cheapest artisan restaurant and its food is delicious!
 4 I don't know what you will look for in terms of price performance point;
 ↳ taste; but yigit restaurant writes a big plus for those who come to work ...
 5
 ↳ Generally good.
 6 What you see is 125 TL in total. It's a pretty convenient place. We can say
 ↳ that it is an artisan restaurant; you should not expect a restaurant; ...
 7
 ↳ Delicious food at rock bottom prices. Friendly staff; Highly recommend
 8 Every time I go, I still experience the amazement I experienced years ago as
 ↳ if it were the first time. There is no need to explain. Folk hero is ...

9
 ↳ The most f/p of all businesses I've seen. ┘

↳ text_clean ┘

0 We went to Marmaris with my wife for a holiday. We chose this restaurant as a
 ↳ place for dinner based on the reviews and because we wanted juicy fo... ┘

1 During my holiday in Marmaris we ate here to fit the food. It's really good. ┘
 ↳ that the food is cheap and nice. Eating as much bread as you want is a... ┘

2 Prices are very affordable. The menu in the photo cost 108 liras. You have to
 ↳ wait 10-15 minutes for food. Staff is annoying. Well it tastes good... ┘

3
 ↳ Turkey's cheapest artisan restaurant and its food is delicious! ┘

4 I don't know what you will look for in terms of price performance point; ┘
 ↳ taste; but yigit restaurant writes a big plus for those who come to work ... ┘

5
 ↳ Generally good. ┘

6 What you see is 125 TL in total. It's a pretty convenient place. We can say
 ↳ that it is an artisan restaurant; you should not expect a restaurant; ... ┘

7
 ↳ Delicious food at rock bottom prices. Friendly staff; Highly recommend ┘

8 Every time I go, I still experience the amazement I experienced years ago as
 ↳ if it were the first time. There is no need to explain. Folk hero is ... ┘

9
 ↳ The most f/p of all businesses I've seen. ┘

1.5.1 3.1 Validate Cleaning

```
[17]: print( 'Empty cleaned reviews:',(model_df['text_clean'].str.len() == 0).sum())
print('Unique original texts:', model_df['text_original'].nunique())
print('Unique cleaned texts:',model_df['text_clean'].nunique())
print('Average words per review:',round(model_df['review_word_count'].mean(),2))
```

```
Empty cleaned reviews: 0
Unique original texts: 1088
Unique cleaned texts: 1088
Average words per review: 20.05
```

```
[18]: short_reviews = (model_df.sort_values('review_word_count')[['business_name',
↳ 'text_clean', 'rating', 'rating_category', 'review_word_count']].head(20))
display(short_reviews)
```

	business_name	text_clean	rating	rating_category \
534	Kasikla Restaurant	Recommend.	5	outdoor_atmosphere
544	Askin Tostevi	Liked.	4	outdoor_atmosphere
500	Kebabci Halil Usta	Expensive.	3	outdoor_atmosphere
582	Kaliteci	Loved.	5	outdoor_atmosphere
424	Bilice	Great.	5	indoor_atmosphere

619	Midyeci Ahmet	Expensive.	3	menu
72	Gold Semaver	Great.	4	outdoor_atmosphere
293	Nazar Cafe	Good.	4	menu
962	Kadineli	Delicious!	5	outdoor_atmosphere
296	Nazar Cafe	Recommend.	4	indoor_atmosphere
314	Sushico	Liked.	4	taste
321	Sardalye	Delicious!	5	indoor_atmosphere
159	Tatbak	Awesomeeee	5	taste
700	Rumeli Iskembecisi	Amazing!	5	outdoor_atmosphere
685	Duveroglu	Liked.	5	outdoor_atmosphere
973	Anzelha	Expensive.	3	indoor_atmosphere
216	Vefa Bozacisi	Perfect!	5	taste
228	Salacak	Great.	4	menu
818	Tavuk Dunyasi	Recommend.	5	outdoor_atmosphere
510	Abidin Tantuni	Deliciousss!	4	taste

	review_word_count
534	1
544	1
500	1
582	1
424	1
619	1
72	1
293	1
962	1
296	1
314	1
321	1
159	1
700	1
685	1
973	1
216	1
228	1
818	1
510	1

1.6 4. Problem Formulation and Technique Selection

Instead of predicting a predetermined result, the client’s objective is to identify recurrent themes in restaurant review text. Labels like ‘slow service,’ ‘cleanliness problem,’ ‘high prices,’ and ‘food quality issue’ are absent from the dataset. Therefore, this is primarily an **unsupervised pattern-discovery problem**.

The analysis will compare two approaches:

1. **TF-IDF + K-Means** as a transparent baseline that groups reviews based on shared words and phrases.

2. **Sentence embeddings + semantic clustering** as a stronger approach that can recognize reviews with similar meanings even when they use different vocabulary.

The clustering features purposefully do not include ratings. They are then used to describe each theme that has been identified as having a tendency toward positive, negative experiences.

A clustering metric by itself will not be used to choose the final model. Theme clarity, cluster separation, representative reviews, and stakeholder utility will all be taken into account.

1.7 5. Baseline Model - TF-IDF+K-Means

K-Means clustering comes after the first clustering method, which uses TF-IDF to represent each review according to its key words and phrases. The top terms in each cluster can be directly examined, making it transparent and easy to inspect. This helps determine whether the resulting groups correspond to significant review themes.

The model is trained on the entire corpus of cleaned reviews, as requested by the client. Ratings are used to interpret the resulting clusters, not as clustering features.

5.1 Build TF-IDF features

```
[19]: tfidf_vectorizer =  
    ↳TfidfVectorizer(lowercase=True, stop_words='english', ngram_range=(1,  
    ↳2), min_df=3, max_df=0.90, max_features=3000)  
X_tfidf = tfidf_vectorizer.fit_transform(model_df['text_clean'])  
print('TF-IDF matrix shape:', X_tfidf.shape)
```

TF-IDF matrix shape: (1100, 957)

5.2 Compare K values

```
[20]: k_values = range(3, 13)  
  
tfidf_inertias = []  
tfidf_silhouettes = []  
  
for k in k_values:  
    model = KMeans(  
        n_clusters=k,  
        random_state=42,  
        n_init=20  
    )  
  
    labels = model.fit_predict(X_tfidf)  
  
    tfidf_inertias.append(  
        model.inertia_  
    )  
  
    tfidf_silhouettes.append(  
        silhouette_score(  
            X_tfidf,  
            labels
```

```

        X_tfidf,
        labels,
        metric='cosine'
    )
)

results_k = pd.DataFrame({
    'K': list(k_values),
    'Inertia': tfidf_inertias,
    'Silhouette': tfidf_silhouettes
})

display(results_k)
#AI-assisted draft

```

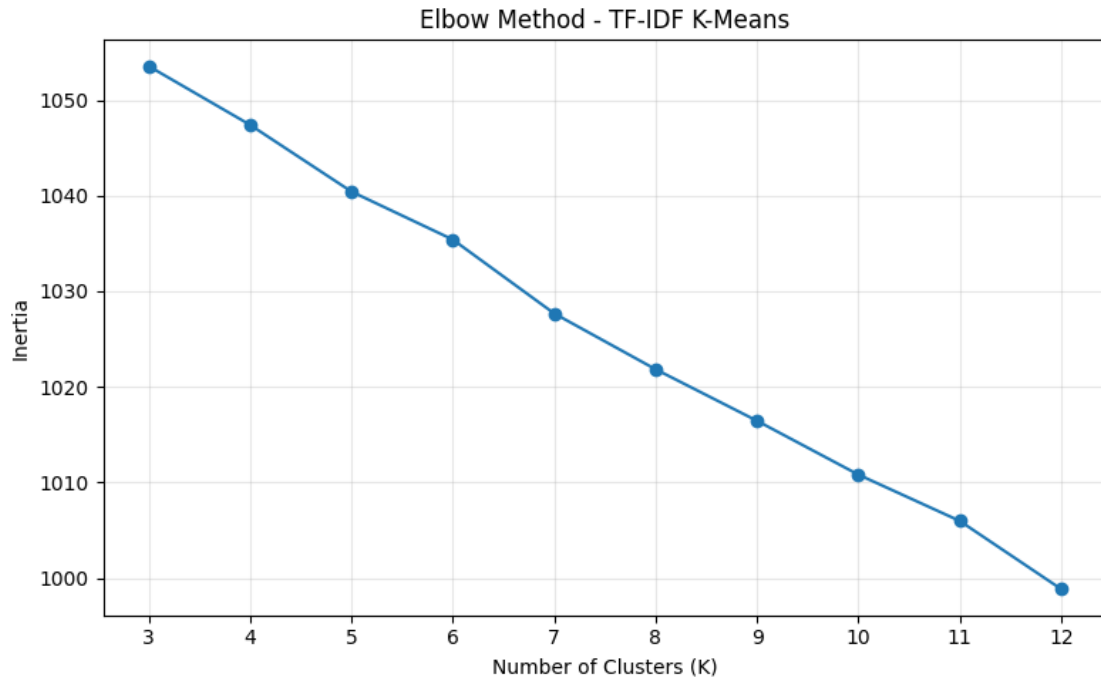
	K	Inertia	Silhouette
0	3	1053.553740	0.018478
1	4	1047.411588	0.018712
2	5	1040.430634	0.022985
3	6	1035.406843	0.023944
4	7	1027.677288	0.026198
5	8	1021.838178	0.030422
6	9	1016.449341	0.030466
7	10	1010.815548	0.032570
8	11	1005.971270	0.038919
9	12	998.844463	0.042166

```

[21]: plt.figure(figsize=(8, 5))
plt.plot( results_k['K'], results_k['Inertia'],marker='o')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Inertia')
plt.title('Elbow Method - TF-IDF K-Means')
plt.xticks(list(k_values))
plt.grid(alpha=0.3)

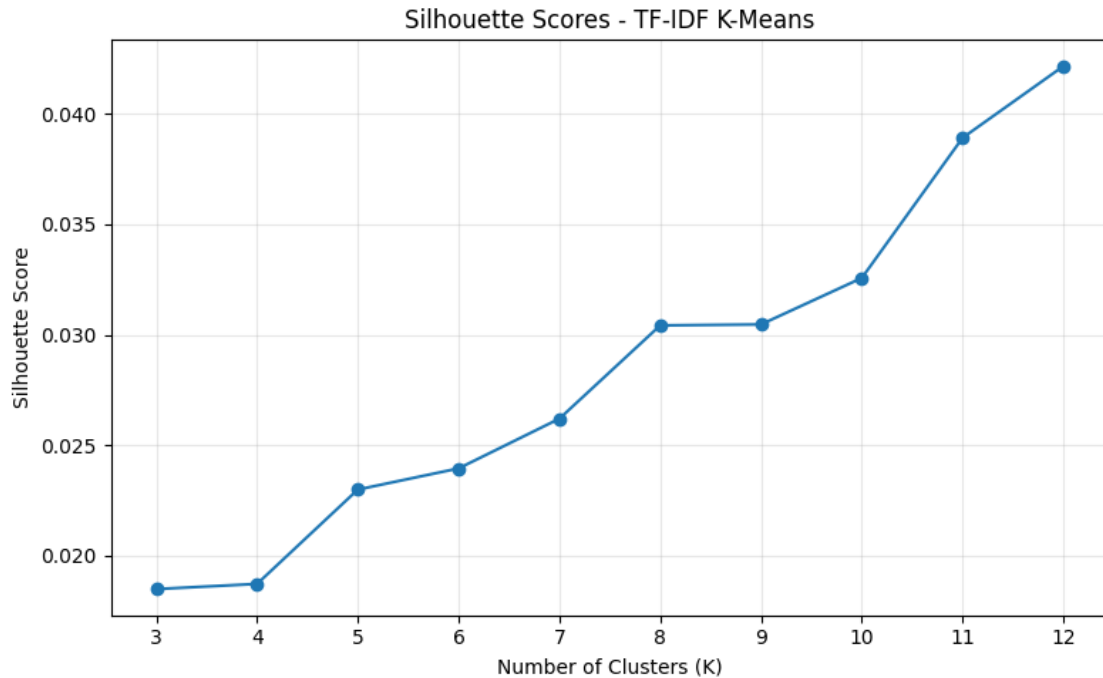
plt.tight_layout()
plt.show()

```

```
[22]: plt.figure(figsize=(8, 5))
plt.plot(results_k['K'], results_k['Silhouette'],marker='o')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Scores - TF-IDF K-Means')
plt.xticks(list(k_values))
plt.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```



5.3 Inspect several candidate K values

```
[23]: candidate_k_values = [5, 6, 7, 8]
```

```
[24]: feature_names = np.array(tfidf_vectorizer.get_feature_names_out())
def inspect_tfidf_clusters(k):
    model = KMeans( n_clusters=k,random_state=42,n_init=20)

    labels = model.fit_predict(X_tfidf)
    model_df[f'tfidf_cluster_{k}'] = labels
    print('-' * 70)
    print(f'K = {k}')
    print('-' * 70)
    for cluster_id in range(k):
        print(f'\nCluster {cluster_id}')
        centroid = model.cluster_centers_[cluster_id]
        top_indices = centroid.argsort()[-10:][::-1]
        top_terms = feature_names[top_indices]
        print('Top terms:', ', '.join(top_terms))
        cluster_reviews = model_df[model_df[f'tfidf_cluster_{k}'] == cluster_id]
        print('Reviews:',len(cluster_reviews))
        display(cluster_reviews[
            ['business_name','text_original', 'rating','rating_category']]).
        head(5) )
```

```
[ ]: for k in candidate_k_values:
      inspect_tfidf_clusters(k)
```

1.7.1 5.4 What We're Looking For

A useful cluster should contain **semantically related terms and reviews that point to a clear, actionable theme**.

For example:

Top terms:

expensive, price, overpriced, money, portion, worth

Representative reviews: - “The food was good but too expensive.” - “Prices are ridiculous for the portion size.” - “Not worth the money.”

This cluster could be confidently interpreted as:

Price / Value Concerns

In contrast, a weak cluster might contain broad terms such as:

good, place, restaurant, nice, food

If its representative reviews discuss several unrelated topics, the cluster would be too broad to provide a meaningful or actionable interpretation.

The goal is therefore not simply to create mathematically distinct clusters, but to identify themes that are **coherent, interpretable, and useful to the stakeholder**.

5.5 Add a cluster profile function

```
[25]: chosen_k_tfidf = 6

tfidf_kmeans = KMeans( n_clusters=chosen_k_tfidf, random_state=42, n_init=20)
model_df['tfidf_cluster'] = ( tfidf_kmeans.fit_predict(X_tfidf))
```

```
[26]: tfidf_cluster_summary = (model_df .groupby('tfidf_cluster').agg(
      reviews=('text_original', 'count'), restaurants=('business_name', '
      ↪ 'unique'), average_rating=('rating', 'mean'), median_rating=('rating', '
      ↪ 'median'), average_words=('review_word_count', 'mean')).round(2))
display(tfidf_cluster_summary)
```

	reviews	restaurants	average_rating	median_rating \
tfidf_cluster				
0	476	99	3.69	4.0
1	189	83	3.75	4.0
2	89	57	4.19	5.0
3	63	45	4.57	5.0
4	145	76	3.99	4.0
5	138	76	4.34	5.0
	average_words			

```
tfidf_cluster
0          21.15
1          20.65
2          20.55
3          13.76
4          19.98
5          18.08
```

```
[27]: tfidf_category_profile = pd.
      ↪ crosstab(model_df['tfidf_cluster'], model_df['rating_category'], normalize='index')
      ↪ * 100
      display(tfidf_category_profile.round(1))
```

rating_category	indoor_atmosphere	menu	outdoor_atmosphere	taste
tfidf_cluster				
0	26.1	18.7	25.2	30.0
1	29.1	15.3	27.5	28.0
2	16.9	20.2	28.1	34.8
3	27.0	15.9	30.2	27.0
4	27.6	21.4	26.9	24.1
5	26.8	13.8	22.5	37.0

```
[ ]: for k in [10, 11, 12]:
      inspect_tfidf_clusters(k)
```

1.7.2 Baseline Model Selection

When the TF-IDF baseline was tested with K values ranging from 3 to 12, the silhouette scores stayed low in all configurations, suggesting that the review text does not naturally split into highly distinct keyword-based clusters. Nevertheless, this baseline's goal goes beyond simple numerical separation. Additionally, the clusters must generate themes that the stakeholder can comprehend and utilize.

K = 11 was chosen as the baseline configuration after representative reviews and top terms were examined. Price concerns, delayed or incorrect orders, poor service, ambiance, and positive recommendation-oriented reviews are some of the relatively obvious themes that emerged from this. When reviews express similar ideas using different vocabulary, TF-IDF may be limited, as evidenced by the fact that some clusters remained broad and mixed. This encourages the next step to test a semantic text representation.

```
[28]: chosen_k_tfidf = 11
      tfidf_kmeans = KMeans( n_clusters=chosen_k_tfidf, random_state=42, n_init=20)
      model_df['tfidf_cluster'] = (tfidf_kmeans.fit_predict(X_tfidf))
```

```
[29]: baseline_summary = ( model_df.groupby('tfidf_cluster') .
      ↪ agg(reviews=('text_original', 'count'), restaurants=('business_name',
      ↪ 'nunique'), average_rating=('rating', 'mean'), median_rating=('rating',
      ↪ 'median'), average_words=('review_word_count', 'mean')).round(2))
      display(baseline_summary)
```

	reviews	restaurants	average_rating	median_rating	\
tfidf_cluster					
0	36	28	4.33	5.0	
1	126	72	4.34	5.0	
2	46	37	3.50	4.0	
3	54	35	3.35	3.0	
4	46	39	2.61	3.0	
5	152	75	3.85	4.0	
6	347	97	3.81	4.0	
7	123	69	4.12	4.0	
8	104	60	4.20	4.0	
9	14	7	3.36	4.0	
10	52	38	4.63	5.0	

	average_words
tfidf_cluster	
0	22.39
1	17.14
2	23.54
3	16.78
4	19.15
5	21.81
6	21.89
7	18.24
8	19.82
9	31.43
10	10.88

```
[30]: baseline_terms = []

for cluster_id in range(chosen_k_tfidf):
    centroid = tfidf_kmeans.cluster_centers_[cluster_id]
    top_indices = centroid.argsort()[-10:][::-1]
    top_terms = feature_names[top_indices]
    baseline_terms.append({'cluster': cluster_id, 'top_terms': ', '.
↪join(top_terms) })
baseline_terms_df = pd.DataFrame( baseline_terms)

display(baseline_terms_df)
```

	cluster	\
0	0	
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	

```

7         7
8         8
9         9
10        10

```

```

                                top_terms
0                                liked, really, really liked, doner,
  place, tried, ate, especially, turkish, good
1                                delicious, food, service, reasonable, food delicious, fresh,
  delicious food, really delicious, place, prices
2                                came, order, late, cold, came late, orders came,
  food came, came cold, orders, order came
3    expensive, expensive place, place, little expensive, little, good, bit
  expensive, prices, taste, expensive restaurant
4                                bad, wasn, service, bad service,
  taste, wasn bad, food, food bad, didn, good
5                                taste, prices, high, amazing, good,
  price, prices high, terms, service, reasonable
6                                recommend, place, pizza,
  excellent, like, service, best, eat, menu, ate
7                                good, service, place, lahmacun, food, ate,
  service good, pretty, good service, fish
8                                nice, place, ambiance, place nice, food, nice
  place, good, nice food, staff, breakfast
9                                ice, ice cream, cream, fethiye, milk,
  road, attention, nice, center, delicious ice
10                               great, thanks, great service, great ambiance, food,
  service, view, great taste, recommend, place

```

1.8 6. Semantic Clustering with Sentence Embeddings

Because similar ideas can be expressed using different vocabulary, some groups remained broad or mixed even tho several useful clusters emerged from the TF-IDF baseline grouping reviews based primarily on shared words and phrases. In order to overcome this constraint, the subsequent model employs sentence embeddings, which represent the review's overall semantic meaning in a dense vector space rather than each review as a sparse wordfrequency vector.

K-Means is once more employed as the clustering algorithm to ensure a fair comparison. This makes it possible to compare the text representation's impact to the TF-IDF baseline.

```
[31]: embedding_model = SentenceTransformer('all-MiniLM-L6-v2')
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

```
Loading weights: 0%|          | 0/103 [00:00<?, ?it/s]
```

6.1 Generate embeddings

```
[32]: review_embeddings = embedding_model.encode(model_df['text_clean']).
      ↪tolist(),show_progress_bar=True,normailze_embeddings=True)
      print('Embedding matrix shape:', review_embeddings.shape)
```

Batches: 0%| | 0/35 [00:00<?, ?it/s]

Embedding matrix shape: (1100, 384)

6.2 Compare K values

```
[33]: semantic_k_values = range(3, 13)
      semantic_inertias = []
      semantic_silhouettes = []

      for k in semantic_k_values:
          model = KMeans(n_clusters=k, random_state=42,n_init=20 )

          labels = model.fit_predict(
              review_embeddings
          )

          semantic_inertias.append(
              model.inertia_
          )

          semantic_silhouettes.append(
              silhouette_score(
                  review_embeddings,
                  labels,
                  metric='cosine'
              )
          )

      semantic_results = pd.DataFrame({'K': list(semantic_k_values), 'Inertia':
      ↪semantic_inertias,'Silhouette': semantic_silhouettes})

      display(semantic_results)
      #AI-assisted draft
```

C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.

warnings.warn(

C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.

warnings.warn(

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

```
warnings.warn(
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
```

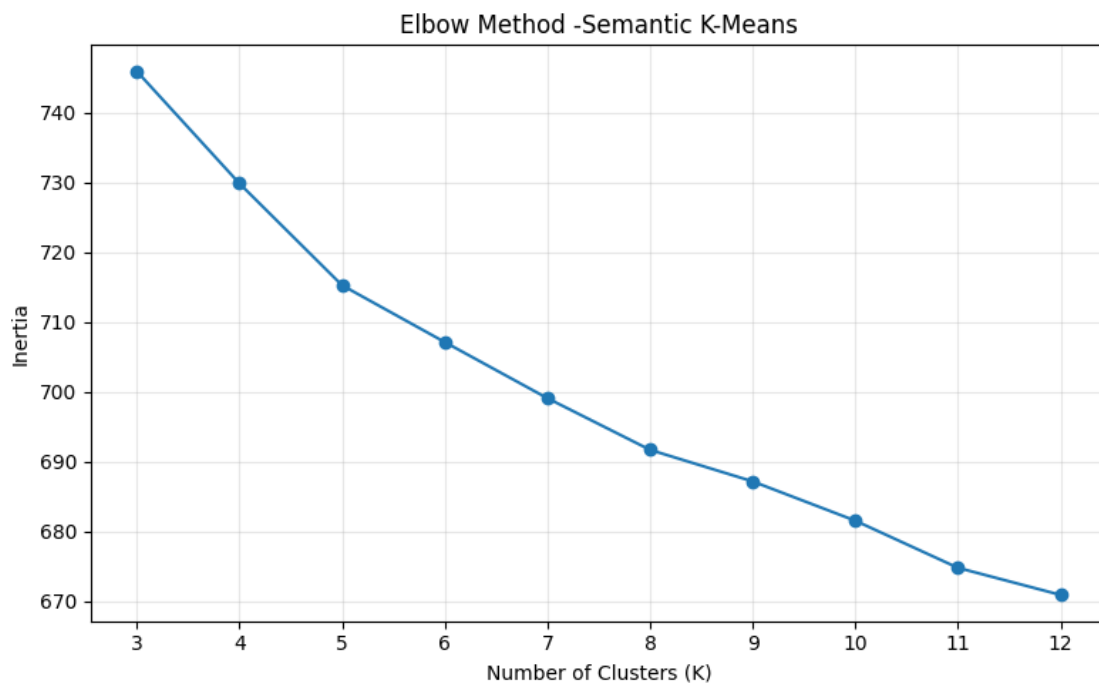
```
warnings.warn(
```

	K	Inertia	Silhouette
0	3	745.895020	0.056726
1	4	729.817627	0.059133
2	5	715.221741	0.068078
3	6	707.084534	0.063283
4	7	699.047729	0.065422
5	8	691.644043	0.064668

6	9	687.122864	0.067407
7	10	681.491760	0.060073
8	11	674.738831	0.063942
9	12	670.854736	0.050753

```
[34]: plt.figure(figsize=(8, 5))
plt.plot(semantic_results['K'], semantic_results['Inertia'], marker='o')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Inertia')
plt.title('Elbow Method -Semantic K-Means')
plt.xticks(list(semantic_k_values))
plt.grid(alpha=0.3)

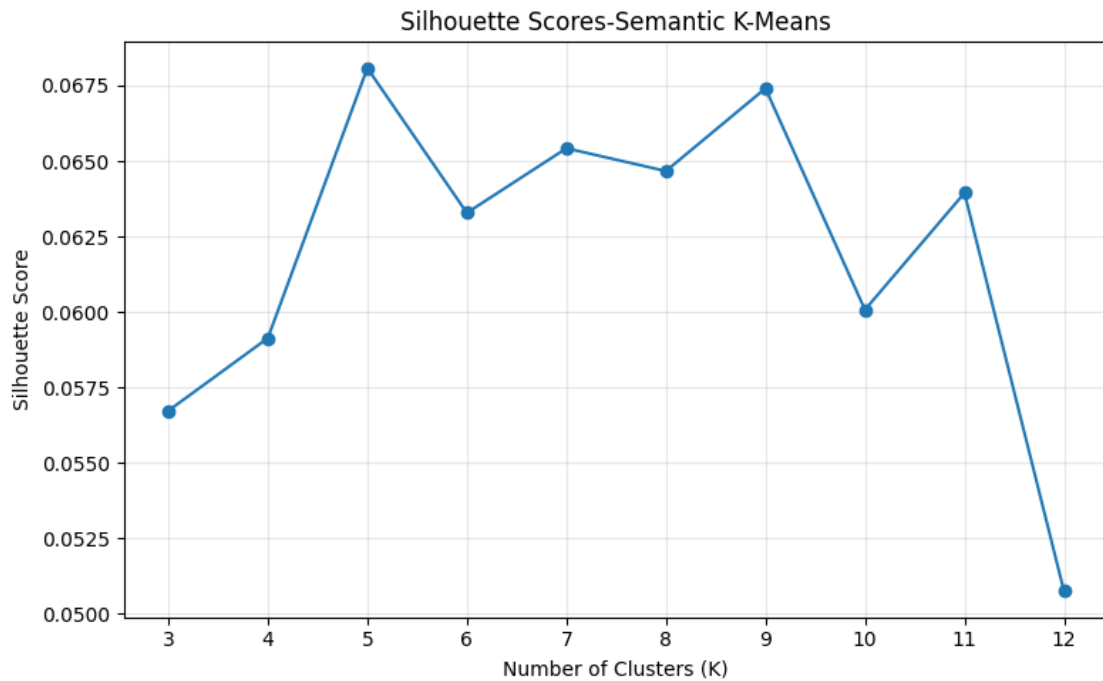
plt.tight_layout()
plt.show()
```



```
[35]: plt.figure(figsize=(8, 5))
plt.plot(semantic_results['K'], semantic_results['Silhouette'], marker='o')

plt.xlabel('Number of Clusters (K)')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Scores-Semantic K-Means')
plt.xticks(list(semantic_k_values))
plt.grid(alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



6.3 Inspect candidate semantic clusters

```
[36]: def inspect_semantic_clusters(k, n_examples=5):
    model = KMeans(n_clusters=k, random_state=42, n_init=20)

    labels = model.fit_predict(
        review_embeddings
    )

    print('=' * 70)
    print(f'Semantic K = {k}')
    print('=' * 70)
    for cluster_id in range(k):
        cluster_indices = np.where(
            labels == cluster_id
        )[0]

        cluster_embeddings = (
            review_embeddings[
                cluster_indices
            ]
        )
```

```

centroid = (
    model.cluster_centers_[
        cluster_id
    ]
    .reshape(1, -1)
)

similarities = cosine_similarity(
    cluster_embeddings,
    centroid
).flatten()

closest_positions = (
    np.argsort(similarities)[::-1]
    [:n_examples]
)

representative_indices = (
    cluster_indices[
        closest_positions
    ]
)

print(
    f'\nCluster {cluster_id}'
)

print(
    'Reviews:',
    len(cluster_indices)
)

display(
    model_df.iloc[
        representative_indices
    ][
        [
            'business_name',
            'text_original',
            'rating',
            'rating_category'
        ]
    ]
)
#AI-assisted draft

```

```
[ ]: for k in [6, 8, 10, 11, 12]:
    inspect_semantic_clusters(k, n_examples=5)
```

1.8.1 Semantic K-Means Results

Sentence embeddings produced more meaningful cluster separation than the TF-IDF baseline-model achieved higher silhouette scores and was better able to group reviews that discussed similar experiences using different vocabulary.

Several clusters began to represent recognizable restaurant experience themes such as service quality, pricing and value, cleanliness, ambiance, food quality, and waiting time.

However, some clusters still combined multiple issues or grouped reviews around specific food types rather than stakeholder relevant review experience themes. Because K-Means forces every review into a cluster, a density-based semantic clustering approach will also be tested before selecting the final model.

```
[37]: hdbscan_settings = [(10, 5),(15, 5),(20, 5),(20, 10),(25, 10),(30, 10)]
      hdbscan_results = []

      for min_cluster_size, min_samples in hdbscan_settings:
          clusterer = hdbscan.HDBSCAN(
              ↪min_cluster_size=min_cluster_size,min_samples=min_samples,metric='euclidean',cluster_select

          labels = clusterer.fit_predict(
              review_embeddings
          )

          n_clusters = len(
              set(labels)
              - {-1}
          )

          noise_points = (
              labels == -1
          ).sum()

          clustered_mask = (
              labels != -1
          )

          if (
              n_clusters > 1
              and clustered_mask.sum() > n_clusters
          ):
              score = silhouette_score(
                  review_embeddings[
                      clustered_mask
```

```

        ],
        labels[
            clustered_mask
        ],
        metric='cosine'
    )
else:
    score = np.nan

    hdbscan_results.append({'min_cluster_size': min_cluster_size, 'min_samples':
↪ min_samples, 'clusters': n_clusters, 'noise': noise_points, 'noise_percent':
↪ noise_points / len(labels) * 100, 'silhouette': score})
hdbscan_results_df = pd.DataFrame(hdbscan_results)
display(hdbscan_results_df)

```

	min_cluster_size	min_samples	clusters	noise	noise_percent	silhouette
0	10	5	0	1100	100.0	NaN
1	15	5	0	1100	100.0	NaN
2	20	5	0	1100	100.0	NaN
3	20	10	0	1100	100.0	NaN
4	25	10	0	1100	100.0	NaN
5	30	10	0	1100	100.0	NaN

1.8.2 Direct HDBSCAN Attempt

When HDBSCAN was first applied directly to the 384 dimensional sentence embeddings across the tested parameter settings, no valid clusters were formed and all 1,100 reviews were categorized as noise. Silhouette scores could not be computed as a result, indicating that it is challenging to directly identify density structure in the high dimensional embedding space. Therefore, before using HDBSCAN once more, a dimensionality-reduction step is tested.

1.9 7. UMAP + HDBSCAN

All reviews were categorized as noise when HDBSCAN was applied directly to the 384-dimensional sentence embeddings. The semantic embeddings are first reduced to a lower dimensional space using UMAP in order to facilitate the identification of density structure.

The reduced representation is then subjected to HDBSCAN, which, in contrast to K-Means, can identify ambiguous reviews as noise and does not force every review into a cluster.

7.1 Reduce the embeddings

```

[38]: umap_model = umap.UMAP( n_neighbors=15, n_components=5,min_dist=0.
↪ 0,metric='cosine',random_state=42)

review_embeddings_umap = umap_model.fit_transform(review_embeddings)
print('Reduced embedding shape:',review_embeddings_umap.shape)

```

C:\Users\tawsi\miniconda3\Lib\site-packages\umap\umap_.py:1952: UserWarning:
n_jobs value 1 overridden to 1 by setting random_state. Use no seed for

```
parallelism.  
    warn(  
  
Reduced embedding shape: (1100, 5)
```

7.2 Test HDBSCAN settings

```
[39]: hdbscan_settings = [  
    (10, 3),  
    (10, 5),  
    (15, 3),  
    (15, 5),  
    (20, 5),  
    (25, 5),  
    (30, 5)  
]  
  
umap_hdbscan_results = []  
  
for min_cluster_size, min_samples in hdbscan_settings:  
  
    clusterer = hdbscan.HDBSCAN(  
        min_cluster_size=min_cluster_size,  
        min_samples=min_samples,  
        metric='euclidean',  
        cluster_selection_method='eom'  
    )  
  
    labels = clusterer.fit_predict(  
        review_embeddings_umap  
    )  
  
    n_clusters = len(  
        set(labels) - {-1}  
    )  
  
    noise_points = (  
        labels == -1  
    ).sum()  
  
    clustered_mask = (  
        labels != -1  
    )  
  
    if (  
        n_clusters > 1  
        and clustered_mask.sum() > n_clusters  
    ):  

```

```

        score = silhouette_score(
            review_embeddings[
                clustered_mask
            ],
            labels[
                clustered_mask
            ],
            metric='cosine'
        )
    else:
        score = np.nan

    umap_hdbscan_results.append({
        'min_cluster_size': min_cluster_size,
        'min_samples': min_samples,
        'clusters': n_clusters,
        'noise': noise_points,
        'noise_percent':
            noise_points / len(labels) * 100,
        'silhouette': score
    })

umap_hdbscan_results_df = pd.DataFrame(
    umap_hdbscan_results
)

display(
    umap_hdbscan_results_df
)

```

	min_cluster_size	min_samples	clusters	noise	noise_percent	silhouette
0	10	3	38	318	28.909091	0.092959
1	10	5	35	335	30.454545	0.092347
2	15	3	20	246	22.363636	0.064551
3	15	5	20	378	34.363636	0.094764
4	20	5	14	278	25.272727	0.040944
5	25	5	11	293	26.636364	0.038841
6	30	5	2	50	4.545455	0.015471

7.3 What a useful result looks like

We are looking for something roughly like:

clusters: 6-15 noise: maybe 5-35%

7.4 Pick one candidate and inspect it

```

[40]: final_hdbscan = hdbscan.HDBSCAN( min_cluster_size=15,min_samples=5,
    ↪metric='euclidean',cluster_selection_method='eom')

```

```

hdbscan_labels = final_hdbscan.fit_predict( review_embeddings_umap)
model_df['hdbscan_cluster'] = (hdbscan_labels)

print(model_df['hdbscan_cluster'].value_counts().sort_index())

```

hdbscan_cluster

```

-1      378
 0       42
 1       26
 2       19
 3       27
 4       33
 5       27
 6       16
 7       20
 8       24
 9       55
10       24
11       33
12       35
13       36
14       23
15       19
16       93
17       32
18       31
19      107

```

Name: count, dtype: int64

7.5 Inspect representative reviews

```
[41]: model_df['hdbscan_probability'] = (final_hdbscan.proBABILITIES_)
```

```

[223]: for cluster_id in sorted(
        model_df['hdbscan_cluster'].unique()
    ):

        if cluster_id == -1:
            continue

        cluster_reviews = (
            model_df[
                model_df['hdbscan_cluster']
                == cluster_id
            ]
            .sort_values(
                'hdbscan_probability',
                ascending=False
            )

```



```

    )
)

print(
    f'\nCluster {cluster_id}'
)

print(
    'Reviews:',
    len(cluster_reviews)
)

display(
    cluster_reviews[
        [ 'business_name', 'text_original', 'rating', 'rating_category',
↪ 'hdbscan_probability' ]
    ].head(5)
)

```

Cluster 0

Reviews: 42

```

    business_name \
14 Pizza Fellas
15 Pizza Fellas
17 Pizza Fellas
18 Pizza Fellas
24     Cafe Inn

```

```

↪                                     text_original \
14 They make delicious pizza with nice ingredients on thin dough. Slice pizza
↪and all pizza options are available. You can try two different flavors ...
15 It's in a nice place but it's very small and their pizza is not that good to
↪be exaggerated. Pizza dough is nice and thin. They do not offer advic...
17 It is very tasty; similar to Italian pizzas. Roast beef and mohair 180 TL;
↪other pizzas 150 TL. An adult male eats 6 slices; pizzas are one size a...
18 Do not come around Marmaris and do not leave without eating pizza at Pizza
↪Fellas. We ate mixed pizza; they can make two types of pizza. So delici...
24 Pizza is delicious.. I can say it is one of the rare tastes. Prices are in
↪the 200-300 range and one pizza is more than enough for two people if y...

```

```

    rating    rating_category  hdbscan_probability
14         5  outdoor_atmosphere                1.0
15         3                taste                1.0
17         4                menu                 1.0
18         5                taste                1.0

```

24 5 indoor_atmosphere 1.0

Cluster 1

Reviews: 26

	business_name	text_original \
5	Haci'nin Yeri - Yigit Lokantasi	Generally good.
72	Gold Semaver	Great.
79	Fethiye Dondurmacisi	Absolutely perfect come try it..
159	Tatbak	Awesomeeee
216	Vefa Bozacisi	Perfect!

	rating	rating_category	hdbscan_probability
5	4	indoor_atmosphere	1.0
72	4	outdoor_atmosphere	1.0
79	5	outdoor_atmosphere	1.0
159	5	taste	1.0
216	5	taste	1.0

Cluster 2

Reviews: 19

	business_name \
53	Sakip Usta Gaziantep
85	Fethiye Dondurmacisi
162	Tatbak
299	Alsancak Firin
300	Alsancak Firin

	text_original \
53	
	Very cheap and delicious!
85	A great taste.
	I went on recommendation; it was really true. I also recommend it to you.
162	
	Delicious!
299	Prices are very affordable; boyoz is fresh and nice. Finding breakfast
	items such as olives; cheese and honey in addition to Boyoz also adds to th...
300	The most beautiful boyoz is clear here! Those who claim otherwise
	should reconsider their taste. I have never eaten such good boyoz in my life.

	rating	rating_category	hdbscan_probability
53	5	menu	1.0
85	5	outdoor_atmosphere	1.0
162	4	taste	1.0
299	5	taste	1.0

300 5 indoor_atmosphere 1.0

Cluster 3

Reviews: 27

	business_name \
49	Sakip Usta Gaziantep
118	Pukka Marmaris
175	Guzel Koy Cafe
257	Cici Tantuni
251	Tuana Restaurant

	text_original \
49	When I ate it in the past; it tasted very good; I think the chef has changed. The atmosphere was average.
118	Great concept and very tasty food.
175	Good quality drinks and food.
257	The pickles that came after you placed your order were very nice. Tantuni is not bad; you can try it. But I would say it's a little oily.
251	They bring delicious yoghurt to the dishes; the portions are hearty; with a beautiful view of the lakeside.

	rating	rating_category	hdbscan_probability
49	2	indoor_atmosphere	1.0
118	5	taste	1.0
175	5	taste	1.0
257	4	taste	1.0
251	4	indoor_atmosphere	1.0

Cluster 4

Reviews: 33

	business_name \
32	Cafe Inn
205	Patso Burger
254	Cici Tantuni
281	Gulcimen Aspava
552	Pizzeria Remo

	text_original \
32	When I went; I waited for almost 20 minutes for the waiters to take care of me. The taste was not bad. The surrounding was beautiful.

Cluster 6

Reviews: 16

	business_name \
54	Sakip Usta Gaziantep
107	Mcdonalds Fethiye
278	Gulcimen Aspava
284	Gulcimen Aspava
780	Sema Gozleme

	text_original \
54	
↪	Ayran came when the meal was over; the food was very bitter.
107	
↪	The potatoes were almost dead and tasteless.
278	The treats were excellent; but I did not like the main dish; it was not fully cooked; other than that; the treats and service were very good.
284	
↪	This is the best aspava restaurant. The treats were perfect.
780	
↪	I didn't like the food very much because it was too salty.

	rating	rating_category	hdbscan_probability
54	5	indoor_atmosphere	1.0
107	2	taste	1.0
278	3	taste	1.0
284	5	indoor_atmosphere	1.0
780	3	outdoor_atmosphere	1.0

Cluster 7

Reviews: 20

	business_name \
148	Leman Kultur
173	Guzel Koy Cafe
239	Nusr-et
358	Bogazkoy Restaurant
930	Sini Mugla

	text_original \
148	
↪	Food was tasteless. The waiters were not smiling at all.
173	
↪	They didn't care about us when we went.

239 It's as if your old meticulousness is gone. The cage arrived cold; the
 ↳salad came for 1; I said two; I did not drink alcohol; but 6 double raki wa...

358

↳ The waiters were understanding; but the food came cold and late.

930 The waiters are scruffy so

↳we didn't feel comfortable eating. The hygiene of the place was also bad.

	rating	rating_category	hdbscan_probability
148	3	taste	1.0
173	1	indoor_atmosphere	1.0
239	2	outdoor_atmosphere	1.0
358	3	indoor_atmosphere	1.0
930	1	indoor_atmosphere	1.0

Cluster 8

Reviews: 24

	business_name \
242	Tuana Restaurant
222	Salacak
325	Sardalye
319	Sardalye
743	Balikci Tanju

	text_original \
242	By the lake; with a beautiful view; ducks are swimming next to you. 1 pan ↳of squid; 1 portion of sea bass; 1 salad; 2 50 sized tuborgs; 2 tea 80 TL.
222	There is a place to eat fish and bread across the beautiful place. If ↳the weather is nice; we go to this doner shop; if not there. You can try.
325	I think they cook very fast and delicious fish; and they have very sweet ↳employees; I definitely recommend them. I also strongly recommend the pic...
319	We ordered sole fish; fried haddock; fried mussels; and sardines. Our ↳favorite among them was sole fish. The only thing that bothered me was that ...

743

↳ Great place and fish; thanks to the amazing chefs

	rating	rating_category	hdbscan_probability
242	4	outdoor_atmosphere	1.0
222	4	outdoor_atmosphere	1.0
325	5	taste	1.0
319	4	outdoor_atmosphere	1.0
743	5	taste	1.0

Cluster 9

Reviews: 55

```

        business_name \
62      Imam Cagdas
720     Antakya Mutfagi
674     Magro
501     Kebabci Halil Usta
390     Metanet Lokantasi

```

```

        text_original \
62  The restaurant that made our four-day Gaziantep holiday poison! There is
    already excessive noise; it is impossible to hear the person next to you...
720
    Thank you very much for bringing the flavors of Antakya here.
674  Magro is a really nice place in terms of design. I ate chicken broth soup;
    tomato soup; magro burger and grilled sea bass. Presentation; portion a...
501
    Deserts was amazing; especially baklava! Liked it.
390  I ate katmer here for the first time and I couldn't
    forget its taste and smell. It may be the most beautiful dessert in the world!!

```

	rating	rating_category	hdbscan_probability
62	1	menu	1.0
720	5	taste	1.0
674	2	menu	1.0
501	5	taste	1.0
390	5	taste	1.0

Cluster 10
Reviews: 24

```

        business_name \
28      Cafe Inn
41      Riviera
168     Guzel Koy Cafe
166     Guzel Koy Cafe
406     Midpoint

```

```

        text_original \
28
    Its breakfast is more satisfying than the places around it.
41  Scrambled eggs with casserole and toast waited 30 minutes. At 12:30pm the
    main menu was denied and showed only breakfast available. Prices are muc...
168
    ordered breakfast; the table was full. It was truly fascinating. I advise.
166
    We went for breakfast in
    the morning; the atmosphere is nice. breakfast items were good and tasty.

```

406 I went on the recommendation of my friend.␣
 ↳Everything at breakfast was delicious. Only the butter had a peculiar smell.

	rating	rating_category	hdbscan_probability
28	4	indoor_atmosphere	1.0
41	2	taste	1.0
168	5	taste	1.0
166	5	indoor_atmosphere	1.0
406	5	taste	1.0

Cluster 11
 Reviews: 33

	business_name \
3	Haci'nin Yeri - Yigit Lokantasi
543	Askin Tostevi
477	Zula
446	Medeniyetler Sofrasi
337	Kumrucu Omur

	text_original \
3	Turkey's cheapest artisan restaurant and its food is delicious!␣
543	Bread; cheese; sausage all left incredible␣ ↳flavors on our palate. We recommend it to everyone who will go to Ayvalık.
477	↳The best hamburger restaurant in Istanbul. Cooking and sauce is very good.␣
446	It is a luxurious place␣ ↳where you can try the delicious tastes of both Turkish and Arabic cuisine.
337	Beautiful place; in a perfect location; clean and equipped with quality;␣ ↳these definitely deserve a star; but as someone who loves dove and has ex...

	rating	rating_category	hdbscan_probability
3	5	indoor_atmosphere	1.0
543	4	menu	1.0
477	5	indoor_atmosphere	1.0
446	5	taste	1.0
337	4	indoor_atmosphere	1.0

Cluster 12
 Reviews: 35

	business_name \
252	Tuana Restaurant
102	Mcdonalds Fethiye
97	My Friends Burger

475 Zula
1001 J Burber

↪ text_original \

252 I recommend the handmade hamburger at a good price; reasonable (50TL) and
↪satisfying. Sprinkled village breakfast (125 TL per person) is a weak; o...

102

↪Taste is exactly the same as in classic hamburgers; prices are extra high.

97 The price was
↪not right; I bought a 2-piece menu; the burger and bread did not taste good.

475 Burgers
↪are great hot dogs do. It is very interesting that french fries are 114 TL.

1001 Hamburger buns are fresh; The patties are soft and
↪delicious. Their special sauces are delicious; It adds a nice flavor to the
↪burger.

	rating	rating_category	hdbscan_probability
252	4	taste	1.0
102	4	menu	1.0
97	2	taste	1.0
475	4	menu	1.0
1001	4	indoor_atmosphere	1.0

Cluster 13
Reviews: 36

business_name \

45 Sakip Usta Gaziantep
52 Sakip Usta Gaziantep
56 Imam Cagdas
64 Imam Cagdas
663 Hasan Usta

↪ text_original \

45 Kebabs are good; trotters are good; desserts are good. Actually; the decor
↪part of the atmosphere is good. Only a lot of noise comes from the kitc...

52

↪We ate 1 mumbar and 1 kusleme and paid 310 TL. The meals were very tasty.

56 We ordered Etli Ali Nazik Kebab.
↪They said he was the most famous. We ate meat with yogurt. The food was cold.

64 I don't know why we ate here despite the exorbitant prices. A portion of
↪kebab or sis meat product starts from 220 TL as of the end of June 2023. ...

663 The space is comfortable and spacious. The service is fast; the staff is
↪concerned. There are many types of kebab. We ordered shish shish and Adan...

	rating	rating_category	hdbscan_probability
45	4	outdoor_atmosphere	1.0
52	4	taste	1.0
56	2	taste	1.0
64	2	indoor_atmosphere	1.0
663	4	indoor_atmosphere	1.0

Cluster 14

Reviews: 23

	business_name \
77	Fethiye Dondurmacisi
182	Cemo Etliemek
237	Nusr-et
274	Bunbun Kas
285	Gulcimen Aspava

	text_original \
77	Bitter; mango; Bodrum tangerine; honey almond; Big
	↳ Babas; mastic gum; walnut; blackberry and forest fruit definitely recommend..
182	I
	↳ had heard that meatloaf was good; but the taste was not that praiseworthy.
237	I do not recommend the lamb cage at all. But skewered and tenderloin are
	↳ great. My carrot slice isn't bad. The waiters are interested and polite; ...
274	We ordered meatballs and handmade meatballs
	↳ that were baked for 6 hours. Meatballs were delicious; meat was delicious.
285	Appetizers and service came quickly. The glasses are badly scratched.
	↳ Especially the fries were fresh and delicious. Raw meatballs and stuffed mea...

	rating	rating_category	hdbscan_probability
77	5	taste	1.0
182	2	outdoor_atmosphere	1.0
237	4	menu	1.0
274	4	menu	1.0
285	3	taste	1.0

Cluster 15

Reviews: 19

	business_name \
108	Mcdonalds Fethiye
323	Sardalye
343	Kofteci Yusuf
454	Backyard
506	Abidin Tantuni

```

                                text_original \
108      I didn't like the gloomy; dark atmosphere of the
environment; the toilets are dirty; you are disgusted to wash your hands.
323      Our back hurt while eating; the place was clean.
343      The place was really dirty and the orders came late.
454 The place is very beautiful; the food is so delicious; you can go with your
pet to one of the rare restaurants that cook the meat to the exact con...
506      It was too oily and the place had problems with cleanliness.

```

	rating	rating_category	hdbscan_probability
108	2	outdoor_atmosphere	1.0
323	2	indoor_atmosphere	1.0
343	3	indoor_atmosphere	1.0
454	5	indoor_atmosphere	1.0
506	2	taste	1.0

Cluster 16
Reviews: 93

```

                                business_name \
2  Haci'nin Yeri - Yigit Lokantasi
39                                Riviera
10 Haci'nin Yeri - Yigit Lokantasi
59                                Imam Cagdas
48                                Sakip Usta Gaziantep

```

```

                                text_original \
2  Prices are very affordable. The menu in the photo cost 108 liras. You have
to wait 10-15 minutes for food. Staff is annoying. Well it tastes good...
39      It was very expensive, but taste wasn't bad.
10      The food was just like the ones my mother made. Prices were reasonable.
59      The taste was very good but the prices were also very high.
48      Taste was amazing! Price was not high.

```

	rating	rating_category	hdbscan_probability
2	3	outdoor_atmosphere	1.0
39	2	menu	1.0
10	5	taste	1.0

59	5	taste	1.0
48	4	menu	1.0

Cluster 17
Reviews: 32

	business_name \
100	Mcdonalds Fethiye
347	Kofteci Yusuf
344	Kofteci Yusuf
241	Nusr-et
459	Backyard

	text_original \
100	I would like to thank the service in Fethiye branch and especially Mr. □ ↳ Samet; the official here, for his interest and relevance.
347	Great service; □ ↳ attention and relevance It was nice. Everything was perfect.
344	Great service. □
241	He's very successful. Even though □ ↳ it was very crowded; the service was incredibly fast.
459	The service provided was very nice; I was happy. □

	rating	rating_category	hdbscan_probability
100	4	taste	1.0
347	5	indoor_atmosphere	1.0
344	4	menu	1.0
241	5	taste	1.0
459	5	outdoor_atmosphere	1.0

Cluster 18
Reviews: 31

	business_name \
37	Riviera
147	Leman Kultur
130	Starbucks Bodrum
170	Guzel Koy Cafe
177	Cemo Etlielmek

	text_original	rating \
37	The staff was very engaged. Service and taste were perfect.	4
147	Good salep was good; the employees were smiling..	4
130	Employees are very warm. in a very pleasant location.	5

170	We were very satisfied from the staff to the meal.	5
177	Employees are kind. Food is delicious	5

	rating_category	hdbscan_probability
37	indoor_atmosphere	1.0
147	indoor_atmosphere	1.0
130	indoor_atmosphere	1.0
170	taste	1.0
177	indoor_atmosphere	1.0

Cluster 19

Reviews: 107

	business_name \
13	Pizza Fellas
22	Cafe Inn
29	Cafe Inn
38	Riviera
43	Riviera

	text_original \
13	It is in the center; easy to reach; close to the parking lot; the prices are partially affordable; the products are delicious but not quite satisf...
22	The environment; the interest and friendliness of the employees; the taste is incredible. It is so beautiful inside that you do not prefer to sit ...
29	Nice intimate restaurant. Good service. Eye for detail. Clean. Really good food; well presented.
38	Perfect location next to the sea. Music; light in the evening lounge atmosphere. In short; everything is beautiful.
43	Nice food and good service; located just near the beach as well.

	rating	rating_category	hdbscan_probability
13	4	indoor_atmosphere	1.0
22	5	indoor_atmosphere	1.0
29	5	outdoor_atmosphere	1.0
38	4	outdoor_atmosphere	1.0
43	5	indoor_atmosphere	1.0

```
[42]: noise_reviews = model_df[ model_df['hdbscan_cluster'] == -1]
print('Noise reviews:',len(noise_reviews))
display( noise_reviews[
    ['business_name', 'text_original', 'rating','rating_category' ]].
    head(10))
```

Noise reviews: 378

	business_name \
0	Haci'nin Yeri - Yigit Lokantasi
4	Haci'nin Yeri - Yigit Lokantasi
6	Haci'nin Yeri - Yigit Lokantasi
8	Haci'nin Yeri - Yigit Lokantasi
9	Haci'nin Yeri - Yigit Lokantasi
11	Pizza Fellas
12	Pizza Fellas
33	Riviera
35	Riviera
36	Riviera

	text_original \
0	We went to Marmaris with my wife for a holiday. We chose this restaurant as a place for dinner based on the reviews and because we wanted juicy fo...
4	I don't know what you will look for in terms of price performance point; taste; but yigit restaurant writes a big plus for those who come to work ...
6	What you see is 125 TL in total. It's a pretty convenient place. We can say that it is an artisan restaurant; you should not expect a restaurant; ...
8	Every time I go, I still experience the amazement I experienced years ago as if it were the first time. There is no need to explain. Folk hero is ...
9	
	The most f/p of all businesses I've seen.
11	The ambiance of the place is nice; when we went; there was a lot of queue; and after waiting for a while; they took our place. The friend who took...
12	Flavor : It has nothing but dough taste. Labor : Not befitting a touristic place. Service : There was no wet wipes and napkins on the table, but t...
33	It is a very, very expensive place. It took 30 minutes for 2 drinks to arrive.
35	quality service and rich menu. the attitude of the staff is really excellent. A business that deserves 5 stars.
36	The sushi was too small so it was not possible to taste the ingredients; the rice was overcooked and lacked flavor; it was missing except for the ...

	rating	rating_category
0	5	taste
4	3	menu
6	5	taste
8	5	outdoor_atmosphere
9	5	indoor_atmosphere
11	5	indoor_atmosphere
12	2	taste
33	3	menu
35	5	outdoor_atmosphere
36	2	taste

1.10 8. Final Model Selection

Three clustering approaches were evaluated:

1. **TF-IDF + K-Means**
2. **SentenceTransformer embeddings + K-Means**
3. **SentenceTransformer embeddings + UMAP + HDBSCAN**

The analysis shows that unsupervised semantic clustering can convert unprocessed restaurant reviews into comprehensible themes that are more valuable than just ratings. Price/value, waiting time, staff/service quality, and cleanliness or environment were the strongest operational patterns. While some themes were heavily concentrated at specific establishments, others were prevalent throughout restaurants.

Additionally, the project demonstrated the significance of model comparison. Semantic embeddings enhanced thematic coherence, TF-IDF offered a transparent baseline, and HDBSCAN identified niche patterns but added significant fragmentation and noise.

All things considered, the final semantic K-Means model offered the best compromise between stakeholder utility, interpretability, and review coverage.

1.10.1 8.1 Final Model Comparison

Cluster separation, review coverage, and qualitative interpretability were used to compare the three methods. Since stakeholder usefulness also depends on whether clusters are coherent, comprehensible, and sufficiently comprehensive, the Silhouette score was treated as one evaluation signal rather than the only selection criterion.

Model	Selected Configuration	Silhouette Score	Review Coverage	Interpretation
TF-IDF + K-Means	K = 11	0.039	100%	Transparent baseline, but several clusters were driven by shared vocabulary and remained broad or mixed
SentenceTransformer + K-Means	K = 11	0.064	100%	Produced more semantically coherent themes while retaining every review
UMAP + HDBSCAN	20 clusters (min_cluster_size=15, min_samples=5)	0.095	65.6%	Identified specific niche themes, but produced substantial fragmentation and classified 34.4% of reviews as noise

Final selection: SentenceTransformer embeddings with K-Means (K = 11) were selected as the main model. While UMAP + HDBSCAN had the highest silhouette score, its greater fragmentation and lower review coverage made it less useful for stakeholder-facing and restaurant-level analysis. As a result, HDBSCAN was kept as a supporting experiment.

```
[43]: final_k = 11
      final_semantic_model = KMeans( n_clusters=final_k, random_state=42,n_init=20)
      model_df['final_cluster'] = (final_semantic_model.
      ↪fit_predict(review_embeddings))
```

```
C:\Users\tawsi\miniconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=5.
  warnings.warn(
```

```
[44]: final_cluster_summary = (model_df.groupby('final_cluster') .
      ↪agg(reviews=('text_original', 'count'),restaurants=('business_name', 'count'),
      ↪'nunique'),average_rating=('rating', 'mean'),median_rating=('rating', 'nunique'),
      ↪'median'),average_words=('review_word_count', 'mean')) .round(2))
      display(final_cluster_summary)
```

	reviews	restaurants	average_rating	median_rating \
final_cluster				
0	86	48	3.72	4.0
1	116	60	4.25	4.0
2	175	76	4.20	5.0
3	128	67	3.75	4.0
4	42	10	4.19	5.0
5	119	66	3.31	3.0
6	78	54	3.27	4.0
7	66	50	3.36	3.0
8	66	31	3.86	4.0
9	153	73	4.27	5.0
10	71	52	4.52	5.0

	average_words
final_cluster	
0	20.42
1	24.60
2	22.48
3	19.51
4	24.50
5	20.87
6	16.24
7	17.47
8	21.29
9	21.84
10	4.75

```
[45]: final_rating_profile = pd.crosstab( model_df['final_cluster'],
      ↪model_df['rating_group'],normalize='index') * 100
      display( final_rating_profile.round(1))
```

rating_group	NEGATIVE (1-2)	MIXED (3)	POSITIVE (4-5)
final_cluster			
0	20.9	14.0	65.1
1	4.3	12.1	83.6
2	8.6	12.6	78.9
3	13.3	21.9	64.8
4	11.9	9.5	78.6
5	31.1	19.3	49.6
6	33.3	14.1	52.6
7	19.7	33.3	47.0

8	10.6	18.2	71.2
9	5.2	12.4	82.4
10	1.4	7.0	91.5

1.11 9. Theme Interpretation and Validation

The final model produced 11 review experience themes, Four of these were identified as operational review experience themes– Price & Value, Waiting Time service Friction, staff & Service Quality, cleanliness, ambience & Environment.

Cluster labels are assigned manually only when the underlying reviews show a sufficiently coherent and understandable theme. Clusters that mix several unrelated ideas are identified as broad or mixed rather than being given an artificially specific label.

```
[46]: from sklearn.metrics.pairwise import cosine_similarity
representative_reviews = {}
for cluster_id in range(final_k):

    cluster_indices = np.where(
        model_df['final_cluster'].values == cluster_id
    )[0]

    cluster_embeddings = review_embeddings[
        cluster_indices
    ]

    centroid = final_semantic_model.cluster_centers_[
        cluster_id
    ].reshape(1, -1)

    similarities = cosine_similarity(
        cluster_embeddings,
        centroid
    ).flatten()

    closest_positions = np.argsort(
        similarities
    )[::-1][:10]

    representative_indices = cluster_indices[
        closest_positions
    ]

    representative_reviews[cluster_id] = (
        model_df.iloc[representative_indices][
            [
                'business_name',
                'text_original',
```

```

        'rating',
        'rating_category'
    ]
]
)

print('*' * 70)
print(f'CLUSTER {cluster_id}')
print('*' * 70)

display(
    representative_reviews[cluster_id]
)

```

CLUSTER 0

```

        business_name \
509      Abidin Tantuni
699  Rumeli Iskembecisi
315          Sushico
1017        Popeyes
508      Abidin Tantuni
320          Sardalye
269        Bunbun Kas
947      Kocak Baklava
769          Umut Pide
61      Imam Cagdas

```

```

        text_original \
509
        The taste is good but not satisfying.
699      The
    taste is excellent; the portion size is good; the price is a bit expensive.
315
        The flavors were good in general; the prices were a bit high.
1017      The
    taste is not bad in general; but you can eat better ones in other restaurants.
508
        it tasted so good that I even ordered and ate the second one!
320  The taste really lives up to expectations. It is a business that maintains
    frying fish; which is essentially a fast food; in accordance with its c...
269  I really liked the handmade basil buttermilk and lemonade. The chicken
    sandwich is okay; I think the chocolate one has nothing to say; it only had...
947      The flavors are still fabulous; Prices may
    change according to the conditions of the day. But the taste is always great.

```

769 I used to come to this
 restaurant a lot. The flavors used to be better. I think the chef has changed.
 61 I had the chance to taste
 many flavors; all of them were very good; I was amazed by Ali Nazik Kebab.

	rating	rating_category
509	4	menu
699	5	taste
315	4	menu
1017	4	taste
508	5	indoor_atmosphere
320	4	taste
269	4	outdoor_atmosphere
947	4	outdoor_atmosphere
769	4	indoor_atmosphere
61	5	indoor_atmosphere

 CLUSTER 1

	business_name \
688	Duveroglu
982	Akpinar Restaurant
753	Yesemek
504	Kebabci Halil Usta
543	Askin Tostevi
548	Askin Tostevi
689	Duveroglu
624	Midyeci Ahmet
219	Vefa Bozacisi
561	Diyarbakir Ocakbasi Palace

	text_original \
688	Good in general. Quality for Kızılay and its surroundings. Lahmacun doesn't taste old...
982	Location view is great. We ate meatballs and kuymak; it was nice. We thought that the dessert and samovar that followed was a treat; but I guess i...
753	The place is nice; Mumbar was smelling bad and the stone came out, I didn't eat it. The mince in the icli kofte was bad; sour. It's also an expensive...
504	We ate küşleme; it was amazing. But as you can see; really really expensive.
543	Bread; cheese; sausage all left incredible flavors on our palate. We recommend it to everyone who will go to Ayvalık.
548	When you go to Ayvalık; I definitely recommend you to go here to eat toast. Perfect taste.

689 I go here for the
↳lahmacun; I haven't tried their other dishes; but the lahmacun is very good.
624 In a word; it is magnificent; this is the heart of mussels in Istanbul.
↳Atom kokoreç must be tasted. The mussels with sauce are another good one..
219
↳ It is a flavor that can be tried when you go to Istanbul.
561 The food is quite delicious. The prices are higher than the average in
↳Samsun; but suitable for a decent place. The waiters were interested; the a...

	rating	rating_category
688	4	menu
982	4	indoor_atmosphere
753	1	taste
504	4	menu
543	4	menu
548	5	taste
689	5	indoor_atmosphere
624	5	indoor_atmosphere
219	4	taste
561	4	menu

CLUSTER 2

	business_name \
870	Classic Thai
996	Kebabci Mahmut
187	Korelee
794	Ciya Sofrasi
50	Sakip Usta Gaziantep
667	Hasan Usta
932	Sini Mugla
678	Magro
238	Nusr-et
787	Asucan Restaurant

↳ text_original \

870 I loved the ambiance; welcome and service. We ate pineapple; spicy
↳shrimp; chicken noodles; and chicken pie. A little pricey but nice place.
996 All the food that came to
↳the table was very high quality and delicious. I recommend it to everyone.
187 The atmosphere of the place and the interest of the waiters were good. Food
↳came fast and hot. We ordered spicy tabokki; it wasn't bitter; it tast..
794 We really liked our orders. There
↳were dishes on the menu that I have never tried in my life; nice variety.

50 We were very satisfied. □
 ↳I strongly recommend trying it. Delicious appetizers come with the meals.
 667 I liked the kebab and salad; but I can't say the same for their appetizers. □
 ↳It was less in variety; unfortunately; it was unpleasant for those who...
 932 Salad was great; appetizers were good. The □
 ↳degree of cooking of the meat was good. Also the toilets were very clean.
 678 Everything I □
 ↳ate that in a super menu; was very good. I recommend this place to everyone.
 238 We've been here many times and each time we left satisfied; Lokum meat; □
 ↳spaghetti; nusret hamburger; salad with tulum cheese; etc. very good but o...
 787 Definitely the chicken and the side salads are also very tasty. Prices are □
 ↳cheap compared to this kind of touristic place. The place was a little ...

	rating	rating_category
870	3	taste
996	5	outdoor_atmosphere
187	4	menu
794	5	taste
50	5	taste
667	4	indoor_atmosphere
932	4	taste
678	5	menu
238	5	taste
787	4	taste

CLUSTER 3

	business_name \
924	Sini Mugla
710	Ege Vera
183	Cemo Etlielmek
10	Haci'nin Yeri - Yigit Lokantasi
429	Baruthane Pilavcisi
642	Monica Kitchen
929	Sini Mugla
1016	Popeyes
623	Midyeci Ahmet
98	My Friends Burger

↳ text_original \ □
 924 □
 ↳Prices are reasonable; foods were delicious.
 710 The prices are reasonable and the food is delicious... The service □
 ↳staff is very caring and attentive...

1095 There are so many types of pizza; you are surprised which one you want to
 taste. I found it very successful in terms of taste. Atmosphere and serv...
 557 It was much better on our last visit (6 years ago).
 This time we were not satisfied with the service or the quality of the pizza.
 24 Pizza is delicious.. I can say it is one of the rare tastes. Prices are in
 the 200-300 range and one pizza is more than enough for two people if y...
 16 I strongly recommend. If I come to the taste part; it was very
 enjoyable to eat with their special sauces. Also the whole pizza was huge.
 1099 Excellent
 pizza and fine wine. Both food and beverage portions are large. I recommend.
 493 The place is spacious; clean. Pizza dough is thin. You had the chance to
 try 6 types of pizza; I can say that the best is the model with spinach a...
 15 It's in a nice place but it's very small and their pizza is not that good
 to be exaggerated. Pizza dough is nice and thin. They do not offer advic...
 1093 Best pizza in the area; very crowded in weekends.
 27 Very enjoyable beautiful place. Pizza salad food taste is great.
 151 Location is nice. We love it as a family. Pizza was delicious.

	rating	rating_category
1095	5	taste
557	3	taste
24	5	indoor_atmosphere
16	5	outdoor_atmosphere
1099	4	outdoor_atmosphere
493	5	taste
15	3	taste
1093	5	outdoor_atmosphere
27	5	taste
151	5	taste

CLUSTER 5

	business_name \
961	Kadineli
717	Antakya Mutfagi
32	Cafe Inn
926	Sini Mugla
890	Inegol Restaurant
439	Baruthane Pilavcisi
974	Anzelha
956	Kocak Baklava
170	Guzel Koy Cafe
1043	Kukis Suadiye

```

                                text_original \
961                                First of all; everything on the menu we tasted
    ↳was quite delicious. The service was fast; the employees were very friendly.
717 The taste of the food was good; but the prices were high according to the
    ↳portion. Waiters tend to take the plate as soon as they eat; so people c...
32                                When I went; I waited for almost 20 minutes for the
    ↳waiters to take care of me. The taste was not bad. The surrounding was
    ↳beautiful.
926 The atmosphere was spacious and the waiters were nice. The meals were
    ↳very delicious... the prices were reasonable; but the portions were small.
890                                The food is
    ↳delicious; but we waited 1 hour after ordering and we were extremely hungry.
439                                The
    ↳menu is very nice; the food is legendary; also the waiters were very polite.
974                                The attention of the
    ↳waiters to us was very nice. In addition; it felt pleasant to eat the hot
    ↳meats.
956                                Even though we went late; the desserts were
    ↳still fresh and delicious. The employees were also very warm and interested.
170                                We were very satisfied from the staff to the meal.
1043 Desserts are good but waiters are very surly; We were not well received.

```

	rating	rating_category
961	5	indoor_atmosphere
717	3	menu
32	2	outdoor_atmosphere
926	4	taste
890	1	taste
439	5	taste
974	4	menu
956	4	taste
170	5	taste
1043	3	taste

```

*****
CLUSTER 6
*****

```

```

                                business_name \
706                                Ege Vera
865                                Mado
459                                Backyard
94  My Friends Burger
455                                Backyard

```


967 Kadineli
 679 Magro
 35 Riviera
 760 Umut Pide
 970 Anzelha

↪ text_original \

706 The service is good; the employees are concerned
 ↪and smiling. Food is fresh delicious. I didn't like the Lahmacun very much.
 865 It may be the most friendly service I have ever encountered in my life. ↪
 ↪All the employees were very sweet; caring people. the drinks and food wer...
 459

↪ The service provided was very nice; I was happy.

94 I never liked the service very bad; when we brought the wrong burger and
 ↪returned it; we had to wait that much longer. The potatoes were terrible;...
 455 The service is slow; I think there is a lack of
 ↪staff and unhappiness; it has not to expensive and it was seedy experience.
 967

↪ Service was great. Recommend and thank you!

679

↪ Great service; great food; everything is great!

35 quality service and rich menu. the
 ↪attitude of the staff is really excellent. A business that deserves 5 stars.
 760 Service was nice; I've tried everything on the menu over time
 ↪and it's all very good. Prices are very cheap; you should definitely come.
 970

↪ Very bad customer satisfaction. The prices are so insane.

	rating	rating_category
706	3	menu
865	5	taste
459	5	outdoor_atmosphere
94	1	taste
455	2	taste
967	4	outdoor_atmosphere
679	5	indoor_atmosphere
35	5	outdoor_atmosphere
760	5	taste
970	1	indoor_atmosphere

 CLUSTER 7

business_name \

51 Sakip Usta Gaziantep
 757 Yesemek

704 Ege Vera
 801 Ciya Sofrasi
 745 Balikci Tanju
 506 Abidin Tantuni
 985 Akpinar Restaurant
 456 Backyard
 896 Karakoy Lokantasi
 661 Hasan Usta

↪ text_original \

51
 ↪The food tastes good. The atmosphere of the place was not very attractive.
 757 The food was delicious but not as exaggerated in other reviews or did not
 ↪appeal to my taste. The environment is quite nice. The place was noisy b...
 704
 ↪The environment is clean and nice. The taste of the food was not bad either.
 801
 ↪ Food was good overall. The toilets are so dirty; we are disgusted.
 745
 ↪ The toilets were very dirty; other than that the food was good.
 506
 ↪ It was too oily and the place had problems with cleanliness.
 985 A restaurant with a beautiful view. It was nice to have
 ↪breakfast with this view. The only problem was that the place was a little
 ↪dirty.
 456
 ↪Although it was crowded; the ambiance and food of the place was nice. Thank
 ↪you.
 896
 ↪ The place was too noisy. We didn't really enjoy the food in this mess.
 661 It is a place that can be
 ↪preferred for taste; but the appetizer variety is very weak. The toilets
 ↪smelled.

	rating	rating_category
51	3	outdoor_atmosphere
757	4	taste
704	3	outdoor_atmosphere
801	3	taste
745	2	menu
506	2	taste
985	4	outdoor_atmosphere
456	4	menu
896	3	taste
661	3	menu

CLUSTER 8

	business_name \
88	My Friends Burger
481	Zula
480	Zula
1010	J Burber
478	Zula
91	My Friends Burger
601	KFC
476	Zula
1007	J Burber
1004	J Burber

	text_original \
88	The hamburger bread was very nice. I didn't like his meatballs very much. Orders are coming out very quickly; and the working friends are very kind...
481	Well prepared hamburger; special taste. But a little bit expensive.here/
480	It was one of the best burgers I've ever eaten. The only downside is that the portion is a little small. Apart from that; it is definitely a place...
1010	Burgers and fries were delicious. Burger meat came well cooked; I suggest you eat it medium cooked. The sauces are quite different and nice; you s...
478	Burgers are super. I've eaten hamburgers in many places; but I've never come across anything like this. Fantastic meal with friendly service in a ...
91	The place is nice; the waiters work sulkily. The location; service and presentation are beautiful. And his hamburger might be the most mediocre ha...
601	The chickens are very nice and tasty; I will not say the same for the hamburger.
476	Hamburger is really delicious. The fries are normal.
1007	The taste of hamburgers is not as good as exaggerated. It's also an expensive restaurant.
1004	The hamburger patties and sauce were very good. I think their mayonnaise is fabricated mayonnaise; It would be tastier if they made it themselves.

	rating	rating_category
88	4	indoor_atmosphere
481	5	outdoor_atmosphere
480	5	outdoor_atmosphere
1010	4	taste
478	5	taste
91	2	taste

601	4	taste
476	5	outdoor_atmosphere
1007	3	outdoor_atmosphere
1004	5	outdoor_atmosphere

CLUSTER 9

	business_name \
604	KFC
846	Salih Usta
894	Karakoy Lokantasi
756	Yesemek
533	Kasikla Restaurant
418	Bilice
143	Leman Kultur
488	Kafros Pizzeria
29	Cafe Inn
934	Sini Mugla

	text_original \
604	The place is very nice; I think, the food was fresh and good; one of my favorite places.
846	it is a very nice place for high quality and special tastes; the employees and the owner are very interested and polite; we are going with love.
894	The location is nice; the menus are varied; the appetizers are delicious; the prices are a little high.
756	It's a great place in terms of taste. The service is fast and the waiters are friendly. It's a bit expensive in terms of price; but you should def...
533	A very pleasant and quiet restaurant. Most importantly; the food is great. Everything I tasted was delicious.
418	The service is absolutely amazing; the working friends are very kind; there is no need to explain the taste; it is a restaurant that should defini...
143	Nice place with a view. Not bad in terms of food but ordinary.
488	I think it's a nice place. Pizza was also nice; but a little expensive. I liked their sauce.
29	Nice intimate restaurant. Good service. Eye for detail. Clean. Really good food; well presented.
934	The satisfying menu of local dishes; the taste of the food and the interest of the employees are very good. The advantage is that it is on the roa...

	rating	rating_category
604	5	taste

846	4	taste
894	4	outdoor_atmosphere
756	5	indoor_atmosphere
533	5	indoor_atmosphere
418	4	outdoor_atmosphere
143	3	taste
488	4	indoor_atmosphere
29	5	outdoor_atmosphere
934	5	indoor_atmosphere

CLUSTER 10

	business_name \
1026	Say Cheese
931	Sini Mugla
332	Kumrucu Omur
655	Sevinc Pastanesi
1041	Kukis Suadiye
600	KFC
544	Askin Tostevi
314	Sushico
685	Duveroglu
85	Fethiye Dondurmacisi

	↳text_original \	
1026	↳ Great taste.	↳
931	↳Great ambiance!	↳
332	↳ I recommend.	Excellent.
655	↳Great ambiance.	↳
1041	Great tastes; the interest of the employees was very good;↳	
	↳thank you to them.	
600	↳was delicious!	It↳
544	↳ Liked.	↳
314	↳ Liked.	↳
685	↳ Liked.	↳
85	A great taste. I went on recommendation; it was really true. I also↳	
	↳recommend it to you.	

	rating	rating_category
1026	3	outdoor_atmosphere
931	5	outdoor_atmosphere
332	5	taste
655	4	indoor_atmosphere
1041	5	outdoor_atmosphere
600	5	outdoor_atmosphere
544	4	outdoor_atmosphere
314	4	taste
685	5	outdoor_atmosphere
85	5	outdoor_atmosphere

9.2 Get keywords for each semantic cluster

```
[47]: cluster_keywords = []

for cluster_id in range(final_k):

    cluster_texts = model_df.loc[
        model_df['final_cluster'] == cluster_id,
        'text_clean'
    ]

    local_vectorizer = TfidfVectorizer(
        stop_words='english',
        ngram_range=(1, 2),
        min_df=2,
        max_features=1000
    )

    try:
        X_local = local_vectorizer.fit_transform(
            cluster_texts
        )

        terms = np.array(
            local_vectorizer.get_feature_names_out()
        )

        mean_scores = np.asarray(
            X_local.mean(axis=0)
        ).flatten()

        top_indices = mean_scores.argsort()[
            -10:
        ][::-1]
```

```

top_terms = terms[top_indices]

cluster_keywords.append({
    'cluster': cluster_id,
    'keywords': ', '.join(top_terms)
})

except ValueError:
    cluster_keywords.append({
        'cluster': cluster_id,
        'keywords': 'Not enough text'
    })

cluster_keywords_df = pd.DataFrame(
    cluster_keywords
)

display(cluster_keywords_df) #AI-assisted draft

```

```

cluster \
0      0
1      1
2      2
3      3
4      4
5      5
6      6
7      7
8      8
9      9
10     10

```

```

↪ keywords
0      taste, good, flavors, coffee, delicious, bad,
↪ cream, ice, ice cream, time
1      place, good, taste, delicious, lahmacun, nice,
↪ eat, food, beautiful, ate
2      good, delicious, ate, place, food, taste,
↪ nice, salad, pasta, soup
3      prices, expensive, delicious, good, high, price, reasonable, prices high,
↪ prices reasonable, taste
4      pizza, nice, pizzas, best, dough, place, taste, best
↪ pizza, good, recommend
5      good, waiters, food, delicious, breakfast, came, service,
↪ taste, place, crowded

```

```

6         service, bad, good, great, business, recommend,
    ↪employees, taste, like, problem
7         place, good, food, nice, ambiance, toilets, taste,
    ↪clean, crowded, atmosphere
8         delicious, good, hamburger, meatballs, meat, burger,
    ↪taste, nice, bread, fries
9         place, nice, good, food, delicious, expensive, great,
    ↪taste, staff, restaurant
10        great, ambiance, recommend, delicious, good, thank,
    ↪nice, bad, liked, perfect

```

9.3 Create the interpretation table

```

[48]: theme_validation = ( final_cluster_summary .reset_index().
    ↪merge(cluster_keywords_df,left_on='final_cluster',right_on='cluster').
    ↪drop(columns='cluster'))
rating_profile_table = ( final_rating_profile.reset_index())
theme_validation = theme_validation.merge(
    ↪rating_profile_table,on='final_cluster')

display(theme_validation)

```

	final_cluster	reviews	restaurants	average_rating	median_rating	\
0	0	86	48	3.72	4.0	
1	1	116	60	4.25	4.0	
2	2	175	76	4.20	5.0	
3	3	128	67	3.75	4.0	
4	4	42	10	4.19	5.0	
5	5	119	66	3.31	3.0	
6	6	78	54	3.27	4.0	
7	7	66	50	3.36	3.0	
8	8	66	31	3.86	4.0	
9	9	153	73	4.27	5.0	
10	10	71	52	4.52	5.0	

	average_words	\
0	20.42	
1	24.60	
2	22.48	
3	19.51	
4	24.50	
5	20.87	
6	16.24	
7	17.47	
8	21.29	
9	21.84	
10	4.75	


```

    keywords \
0      taste, good, flavors, coffee, delicious, bad,
    cream, ice, ice cream, time
1      place, good, taste, delicious, lahmacun, nice,
    eat, food, beautiful, ate
2      good, delicious, ate, place, food, taste,
    nice, salad, pasta, soup
3      prices, expensive, delicious, good, high, price, reasonable, prices high,
    prices reasonable, taste
4      pizza, nice, pizzas, best, dough, place, taste, best
    pizza, good, recommend
5      good, waiters, food, delicious, breakfast, came, service,
    taste, place, crowded
6      service, bad, good, great, business, recommend,
    employees, taste, like, problem
7      place, good, food, nice, ambiance, toilets, taste,
    clean, crowded, atmosphere
8      delicious, good, hamburger, meatballs, meat, burger,
    taste, nice, bread, fries
9      place, nice, good, food, delicious, expensive, great,
    taste, staff, restaurant
10     great, ambiance, recommend, delicious, good, thank,
    nice, bad, liked, perfect

```

	NEGATIVE (1-2)	MIXED (3)	POSITIVE (4-5)
0	20.930233	13.953488	65.116279
1	4.310345	12.068966	83.620690
2	8.571429	12.571429	78.857143
3	13.281250	21.875000	64.843750
4	11.904762	9.523810	78.571429
5	31.092437	19.327731	49.579832
6	33.333333	14.102564	52.564103
7	19.696970	33.333333	46.969697
8	10.606061	18.181818	71.212121
9	5.228758	12.418301	82.352941
10	1.408451	7.042254	91.549296

```

[49]: theme_labels = {0: 'Food Taste & Satisfaction', 1: 'Regional / Specialty Food
    Experience', 2: 'Overall Dining Experience & Menu Variety', 3: 'Price &
    Value', 4: 'Pizza Experience', 5: 'Waiting Time & Service Friction', 6: 'Staff
    & Service Quality', 7: 'Cleanliness, Ambience & Environment', 8: 'Burger
    Experience', 9: 'Positive Overall Experience', 10: 'Short Positive Praise /
    Recommendation'}

```

```

theme_confidence = {0: 'Medium',1: 'Medium',2: 'Medium',3: 'High',4: 'High',5: 'High',6: 'High',7: 'High',8: 'High', 9: 'Medium',10: 'High'}
model_df['theme'] = (model_df['final_cluster'].map(theme_labels))
model_df['theme_confidence'] = ( model_df['final_cluster'] .
    map(theme_confidence))

```

```

[50]: final_theme_table = ( model_df.groupby(
    ['final_cluster','theme','theme_confidence' ] ).
    agg(reviews=('text_original', 'count'),restaurants=('business_name', 'nunique'),
    average_rating=('rating', 'mean'),median_rating=('rating', 'median')) .reset_index()

final_theme_table['restaurant_coverage_percent'] = (
    final_theme_table['restaurants'] / model_df['business_name'].nunique()* 100)
final_theme_table[ 'average_rating'] = ( final_theme_table['average_rating'].
    round(2))
final_theme_table[ 'restaurant_coverage_percent'] = (final_theme_table[
    'restaurant_coverage_percent'].round(1))
display(final_theme_table)

```

	final_cluster	theme	theme_confidence \
0	0	Food Taste & Satisfaction	Medium
1	1	Regional / Specialty Food Experience	Medium
2	2	Overall Dining Experience & Menu Variety	Medium
3	3	Price & Value	High
4	4	Pizza Experience	High
5	5	Waiting Time & Service Friction	High
6	6	Staff & Service Quality	High
7	7	Cleanliness, Ambience & Environment	High
8	8	Burger Experience	High
9	9	Positive Overall Experience	Medium
10	10	Short Positive Praise / Recommendation	High

	reviews	restaurants	average_rating	median_rating \
0	86	48	3.72	4.0
1	116	60	4.25	4.0
2	175	76	4.20	5.0
3	128	67	3.75	4.0
4	42	10	4.19	5.0
5	119	66	3.31	3.0
6	78	54	3.27	4.0
7	66	50	3.36	3.0
8	66	31	3.86	4.0
9	153	73	4.27	5.0
10	71	52	4.52	5.0

restaurant_coverage_percent

0	48.0
1	60.0
2	76.0
3	67.0
4	10.0
5	66.0
6	54.0
7	50.0
8	31.0
9	73.0
10	52.0

```
[51]: restaurant_theme_counts = pd.crosstab(model_df['business_name'],
      ↪model_df['theme'])
restaurant_theme_percent = ( restaurant_theme_counts.div(
      ↪restaurant_theme_counts.sum(axis=1),axis=0)* 100)
display( restaurant_theme_percent.head())
```

theme	Burger Experience	Cleanliness, Ambience & Environment \
business_name		
Abidin Tantuni	0.000000	9.090909
Ahsap Iskender	9.090909	18.181818
Akpinar Restaurant	18.181818	18.181818
Alavara	0.000000	0.000000
Alsancak Firin	0.000000	0.000000

theme	Food Taste & Satisfaction \
business_name	
Abidin Tantuni	36.363636
Ahsap Iskender	9.090909
Akpinar Restaurant	0.000000
Alavara	0.000000
Alsancak Firin	54.545455

theme	Overall Dining Experience & Menu Variety \
business_name	
Abidin Tantuni	0.000000
Ahsap Iskender	9.090909
Akpinar Restaurant	0.000000
Alavara	63.636364
Alsancak Firin	27.272727

theme	Pizza Experience	Positive Overall Experience \
business_name		
Abidin Tantuni	0.0	0.000000
Ahsap Iskender	0.0	0.000000
Akpinar Restaurant	0.0	9.090909
Alavara	0.0	18.181818

Alsancak Firin	0.0	0.000000
----------------	-----	----------

theme	Price & Value	Regional / Specialty Food Experience \
business_name		
Abidin Tantuni	9.090909	27.272727
Ahsap Iskender	27.272727	9.090909
Akpinar Restaurant	9.090909	9.090909
Alavara	9.090909	0.000000
Alsancak Firin	18.181818	0.000000

theme	Short Positive Praise / Recommendation \
business_name	
Abidin Tantuni	18.181818
Ahsap Iskender	0.000000
Akpinar Restaurant	9.090909
Alavara	0.000000
Alsancak Firin	0.000000

theme	Staff & Service Quality	Waiting Time & Service Friction
business_name		
Abidin Tantuni	0.000000	0.000000
Ahsap Iskender	9.090909	9.090909
Akpinar Restaurant	9.090909	18.181818
Alavara	0.000000	9.090909
Alsancak Firin	0.000000	0.000000

```
[52]: top_restaurants_by_theme = []

for theme in restaurant_theme_percent.columns:

    top_restaurants = (
        restaurant_theme_percent[theme]
        .sort_values(ascending=False)
        .head(5)
    )

    for restaurant, percentage in top_restaurants.items():
        count = restaurant_theme_counts.loc[
            restaurant,
            theme
        ]

        top_restaurants_by_theme.append({
            'theme': theme,
            'restaurant': restaurant,
            'reviews_in_theme': count,
            'theme_share_percent': round(
```

```

        percentage,
        1
    )
})

top_restaurants_by_theme_df = pd.DataFrame(
    top_restaurants_by_theme
)

display(
    top_restaurants_by_theme_df
)#AI-assisted draft

```

	theme	restaurant \
0	Burger Experience	J Burber
1	Burger Experience	Zula
2	Burger Experience	Bunbun Kas
3	Burger Experience	KFC
4	Burger Experience	My Friends Burger
5	Cleanliness, Ambience & Environment	Mcdonalds Fethiye
6	Cleanliness, Ambience & Environment	Ahsap Iskender
7	Cleanliness, Ambience & Environment	Berdan Doner
8	Cleanliness, Ambience & Environment	Akpinar Restaurant
9	Cleanliness, Ambience & Environment	Guzel Koy Cafe
10	Food Taste & Satisfaction	Alsancak Firin
11	Food Taste & Satisfaction	Suflor
12	Food Taste & Satisfaction	Fethiye Dondurmacisi
13	Food Taste & Satisfaction	Abidin Tantuni
14	Food Taste & Satisfaction	Kruvasan
15	Overall Dining Experience & Menu Variety	Alavara
16	Overall Dining Experience & Menu Variety	Say Cheese
17	Overall Dining Experience & Menu Variety	Sushico
18	Overall Dining Experience & Menu Variety	Antakya Mutfagi
19	Overall Dining Experience & Menu Variety	Kumsal Lokantasi
20	Pizza Experience	Miss Pizza
21	Pizza Experience	Kafros Pizzeria
22	Pizza Experience	Pizzeria Remo
23	Pizza Experience	Pizza Fellas
24	Pizza Experience	Cafe Inn
25	Positive Overall Experience	Churchill Bistro
26	Positive Overall Experience	Minoa
27	Positive Overall Experience	Leman Kultur
28	Positive Overall Experience	Backyard
29	Positive Overall Experience	Lipa
30	Price & Value	Haci'nin Yeri - Yigit Lokantasi
31	Price & Value	Kebabci Mahmut
32	Price & Value	Imam Cagdas
33	Price & Value	Baskent Aspava

34	Price & Value	Ege Vera
35	Regional / Specialty Food Experience	Metanet Lokantasi
36	Regional / Specialty Food Experience	Vefa Bozacisi
37	Regional / Specialty Food Experience	Midyeci Ahmet
38	Regional / Specialty Food Experience	Diyarbakir Ocakbasi Palace
39	Regional / Specialty Food Experience	Kocak Baklava
40	Short Positive Praise / Recommendation	Cici Tantuni
41	Short Positive Praise / Recommendation	Gold Semaver
42	Short Positive Praise / Recommendation	Tatbak
43	Short Positive Praise / Recommendation	Abidin Tantuni
44	Short Positive Praise / Recommendation	Fethiye Dondurmacisi
45	Staff & Service Quality	Flash
46	Staff & Service Quality	Riviera
47	Staff & Service Quality	Medeniyetler Sofrasi
48	Staff & Service Quality	Koco Restaurant
49	Staff & Service Quality	Diyarbakir Ocakbasi Palace
50	Waiting Time & Service Friction	Nesh
51	Waiting Time & Service Friction	Gulcimen Aspava
52	Waiting Time & Service Friction	Kruvasan
53	Waiting Time & Service Friction	Tavuk Dunyasi
54	Waiting Time & Service Friction	Kofteci Yusuf

	reviews_in_theme	theme_share_percent
0	10	90.9
1	8	72.7
2	4	36.4
3	4	36.4
4	3	27.3
5	4	36.4
6	2	18.2
7	2	18.2
8	2	18.2
9	2	18.2
10	6	54.5
11	5	45.5
12	4	36.4
13	4	36.4
14	4	36.4
15	7	63.6
16	6	54.5
17	6	54.5
18	5	45.5
19	5	45.5
20	9	81.8
21	8	72.7
22	8	72.7
23	7	63.6
24	4	36.4

25	8	72.7
26	6	54.5
27	6	54.5
28	5	45.5
29	5	45.5
30	6	54.5
31	5	45.5
32	4	36.4
33	4	36.4
34	3	27.3
35	9	81.8
36	6	54.5
37	5	45.5
38	4	36.4
39	4	36.4
40	3	27.3
41	3	27.3
42	3	27.3
43	2	18.2
44	2	18.2
45	3	27.3
46	3	27.3
47	3	27.3
48	3	27.3
49	2	18.2
50	7	63.6
51	6	54.5
52	5	45.5
53	5	45.5
54	4	36.4

```
[53]: top_restaurants_by_theme_df[ 'evidence_level'] = np.where(
      top_restaurants_by_theme_df['reviews_in_theme'] >= 3, 'Stronger', 'Limited')
      display( top_restaurants_by_theme_df)
```

	theme	restaurant \
0	Burger Experience	J Burber
1	Burger Experience	Zula
2	Burger Experience	Bunbun Kas
3	Burger Experience	KFC
4	Burger Experience	My Friends Burger
5	Cleanliness, Ambience & Environment	Mcdonalds Fethiye
6	Cleanliness, Ambience & Environment	Ahsap Iskender
7	Cleanliness, Ambience & Environment	Berdan Doner
8	Cleanliness, Ambience & Environment	Akpinar Restaurant
9	Cleanliness, Ambience & Environment	Guzel Koy Cafe
10	Food Taste & Satisfaction	Alsancak Firin
11	Food Taste & Satisfaction	Suflor

12	Food Taste & Satisfaction	Fethiye Dondurmacisi
13	Food Taste & Satisfaction	Abidin Tantuni
14	Food Taste & Satisfaction	Kruvasan
15	Overall Dining Experience & Menu Variety	Alavara
16	Overall Dining Experience & Menu Variety	Say Cheese
17	Overall Dining Experience & Menu Variety	Sushico
18	Overall Dining Experience & Menu Variety	Antakya Mutfagi
19	Overall Dining Experience & Menu Variety	Kumsal Lokantasi
20	Pizza Experience	Miss Pizza
21	Pizza Experience	Kafros Pizzeria
22	Pizza Experience	Pizzeria Remo
23	Pizza Experience	Pizza Fellas
24	Pizza Experience	Cafe Inn
25	Positive Overall Experience	Churchill Bistro
26	Positive Overall Experience	Minoa
27	Positive Overall Experience	Leman Kultur
28	Positive Overall Experience	Backyard
29	Positive Overall Experience	Lipa
30	Price & Value	Haci'nin Yeri - Yigit Lokantasi
31	Price & Value	Kebabci Mahmut
32	Price & Value	Imam Cagdas
33	Price & Value	Baskent Aspava
34	Price & Value	Ege Vera
35	Regional / Specialty Food Experience	Metanet Lokantasi
36	Regional / Specialty Food Experience	Vefa Bozacisi
37	Regional / Specialty Food Experience	Midyeci Ahmet
38	Regional / Specialty Food Experience	Diyarbakir Ocakbasi Palace
39	Regional / Specialty Food Experience	Kocak Baklava
40	Short Positive Praise / Recommendation	Cici Tantuni
41	Short Positive Praise / Recommendation	Gold Semaver
42	Short Positive Praise / Recommendation	Tatbak
43	Short Positive Praise / Recommendation	Abidin Tantuni
44	Short Positive Praise / Recommendation	Fethiye Dondurmacisi
45	Staff & Service Quality	Flash
46	Staff & Service Quality	Riviera
47	Staff & Service Quality	Medeniyetler Sofrasi
48	Staff & Service Quality	Koco Restaurant
49	Staff & Service Quality	Diyarbakir Ocakbasi Palace
50	Waiting Time & Service Friction	Nesh
51	Waiting Time & Service Friction	Gulcimen Aspava
52	Waiting Time & Service Friction	Kruvasan
53	Waiting Time & Service Friction	Tavuk Dunyasi
54	Waiting Time & Service Friction	Kofteci Yusuf

	reviews_in_theme	theme_share_percent	evidence_level
0	10	90.9	Stronger
1	8	72.7	Stronger
2	4	36.4	Stronger

3	4	36.4	Stronger
4	3	27.3	Stronger
5	4	36.4	Stronger
6	2	18.2	Limited
7	2	18.2	Limited
8	2	18.2	Limited
9	2	18.2	Limited
10	6	54.5	Stronger
11	5	45.5	Stronger
12	4	36.4	Stronger
13	4	36.4	Stronger
14	4	36.4	Stronger
15	7	63.6	Stronger
16	6	54.5	Stronger
17	6	54.5	Stronger
18	5	45.5	Stronger
19	5	45.5	Stronger
20	9	81.8	Stronger
21	8	72.7	Stronger
22	8	72.7	Stronger
23	7	63.6	Stronger
24	4	36.4	Stronger
25	8	72.7	Stronger
26	6	54.5	Stronger
27	6	54.5	Stronger
28	5	45.5	Stronger
29	5	45.5	Stronger
30	6	54.5	Stronger
31	5	45.5	Stronger
32	4	36.4	Stronger
33	4	36.4	Stronger
34	3	27.3	Stronger
35	9	81.8	Stronger
36	6	54.5	Stronger
37	5	45.5	Stronger
38	4	36.4	Stronger
39	4	36.4	Stronger
40	3	27.3	Stronger
41	3	27.3	Stronger
42	3	27.3	Stronger
43	2	18.2	Limited
44	2	18.2	Limited
45	3	27.3	Stronger
46	3	27.3	Stronger
47	3	27.3	Stronger
48	3	27.3	Stronger
49	2	18.2	Limited
50	7	63.6	Stronger

51	6	54.5	Stronger
52	5	45.5	Stronger
53	5	45.5	Stronger
54	4	36.4	Stronger

```
[54]: operational_themes = ['Price & Value','Waiting Time & Service Friction','Staff_
    ↳& Service Quality','Cleanliness, Ambience & Environment']

operational_heatmap = (
    restaurant_theme_percent[
        operational_themes
    ]
    .copy()
)

operational_heatmap = operational_heatmap[
    operational_heatmap.max(axis=1) >= 18.2
]

# sort by strongest operational theme
operational_heatmap['max_theme'] = (operational_heatmap.max(axis=1))
operational_heatmap = (operational_heatmap.
    ↳sort_values('max_theme',ascending=False).drop(columns='max_theme'))
display(operational_heatmap.head())
```

theme	Price & Value \
business_name	
Nesh	0.000000
Haci'nin Yeri - Yigit Lokantasi	54.545455
Gulcimen Aspava	0.000000
Kruvasan	0.000000
Kebabci Mahmut	45.454545

theme	Waiting Time & Service Friction \
business_name	
Nesh	63.636364
Haci'nin Yeri - Yigit Lokantasi	0.000000
Gulcimen Aspava	54.545455
Kruvasan	45.454545
Kebabci Mahmut	0.000000

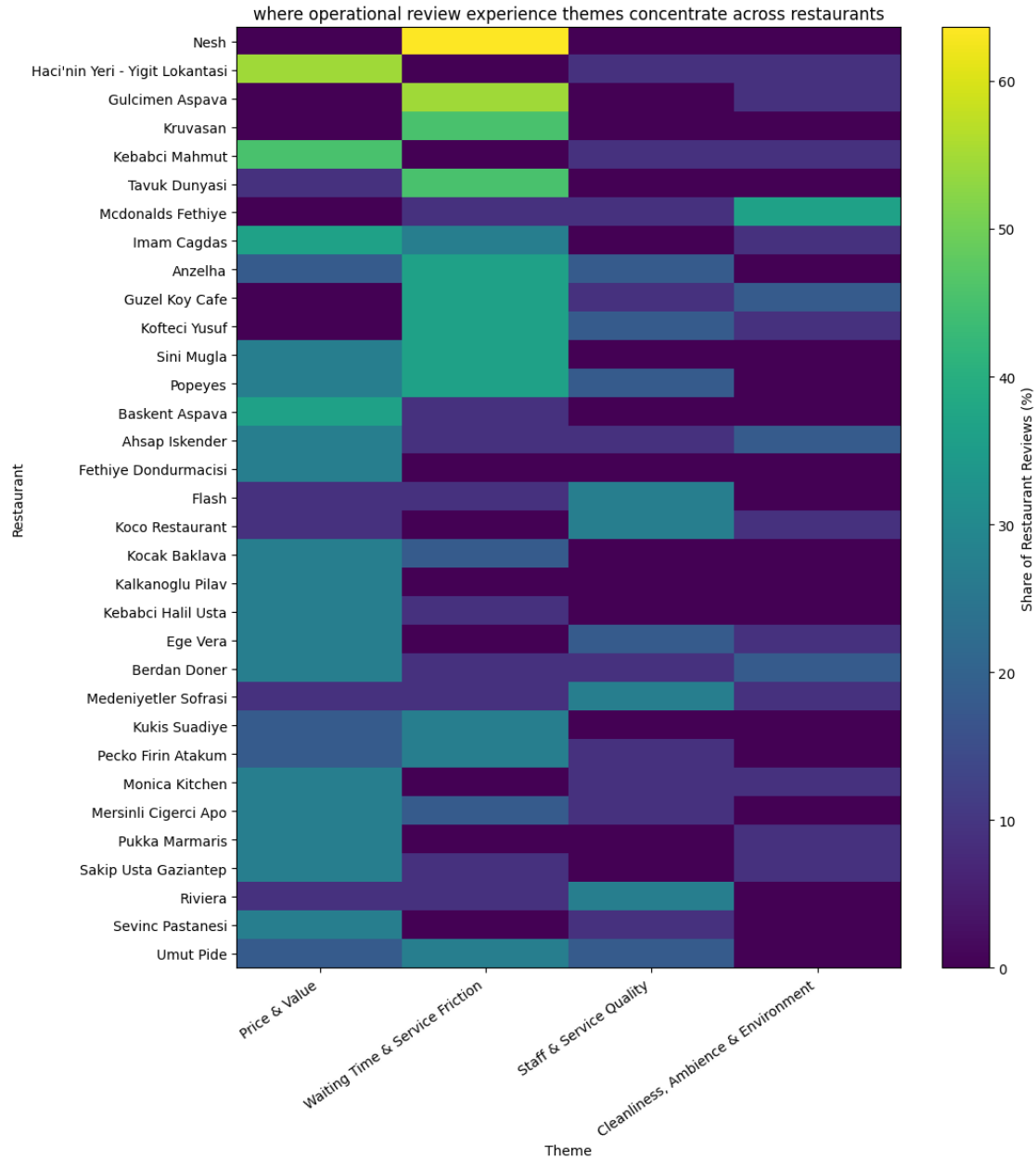
theme	Staff & Service Quality \
business_name	
Nesh	0.000000
Haci'nin Yeri - Yigit Lokantasi	9.090909
Gulcimen Aspava	0.000000
Kruvasan	0.000000
Kebabci Mahmut	9.090909

theme	Cleanliness, Ambience & Environment
business_name	
Nesh	0.000000
Haci'nin Yeri - Yigit Lokantasi	9.090909
Gulcimen Aspava	9.090909
Kruvasan	0.000000
Kebabci Mahmut	9.090909

```
[55]: plt.figure(figsize=(11, 12))
plt.imshow(operational_heatmap, aspect='auto')
plt.colorbar( label='Share of Restaurant Reviews (%)')
plt.xticks(range(len(operational_heatmap.columns)), operational_heatmap.
    ↪columns, rotation=35, ha='right')

plt.yticks(range(len(operational_heatmap.index)), operational_heatmap.index)
plt.title('where operational review experience themes concentrate across_
    ↪restaurants')
plt.xlabel('Theme')
plt.ylabel('Restaurant')

plt.tight_layout()
plt.show()
```



1.11.1 Cross-Restaurant Pattern

The heatmap demonstrates that operational review experience themes are not evenly distributed among restaurants; some are closely linked to a single, predominant issue, such as cleanliness, waiting times, or prices, while others exhibit a more evenly distributed range of issues.

This implies that some review experience themes might represent operational issues unique to restaurants rather than problems that affect the entire industry. Nevertheless, the percentages should be interpreted in conjunction with the underlying review counts rather than as exact population estimates because each restaurant has only 11 reviews.

1.12 10. Cross-Restaurant Theme Comparison

In order to determine whether the final themes were widely dispersed or concentrated at specific locations, they were compared across restaurants. There are only 11 reviews for each restaurant in the dataset, so restaurant-level percentages need to be interpreted carefully. A single review makes up about 9.1% of a restaurant's available sample. Because of this, percentages and raw review counts are both provided.

Smaller counts are considered limited evidence, whereas operational themes with at least three supporting reviews at a restaurant are considered stronger evidence.

1.13 11. Restaurant-Type Comparison

The client was especially curious about whether different restaurant types had different review experience themes.

Since there is no explicit restaurant-type field in the dataset, restaurant type is only inferred conservatively when the business name includes a clear type indicator, like 'Pizza,' 'Burger,' 'Cafe,' or 'Kebab.' Restaurants classified as 'Other / Unknown' are those whose type cannot be reliably determined from their name.

This derived field is not regarded as verified restaurant metadata and is only utilized for exploratory comparison.

```
[56]: def derive_restaurant_type(name):
      name = str(name).lower()

      if any(word in name for word in ['pizza', 'pizzeria']):
          return 'Pizza'

      if any(word in name for word in ['burger', 'burber']):
          return 'Burger'

      if 'cafe' in name:
          return 'Cafe'

      if any(word in name for word in ['kebab', 'kebabci']):
          return 'Kebab'

      if any(word in name for word in ['dondurma', 'dondurmacisi']):
          return 'Dessert / Ice Cream'

      if any(word in name for word in ['pastanesi', 'firin', 'baklava']):
          return 'Bakery / Dessert'

      if any(word in name for word in ['lokantasi', 'lokantası']):
          return 'Local Restaurant'

      if 'restaurant' in name:
          return 'Restaurant'
```

```

    return 'Other / Unknown'
model_df['restaurant_type'] = (
    model_df['business_name']
    .apply(derive_restaurant_type)
)

```

```

[57]: restaurant_type_summary = (
    model_df[
        ['business_name', 'restaurant_type']
    ]
    .drop_duplicates()
    ['restaurant_type']
    .value_counts()
)

display(
    restaurant_type_summary
    .to_frame('restaurants')
)

```

restaurants	
restaurant_type	
Other / Unknown	70
Restaurant	8
Local Restaurant	5
Pizza	4
Bakery / Dessert	4
Cafe	3
Burger	3
Kebab	2
Dessert / Ice Cream	1

```

[58]: restaurant_type_mapping = ( model_df[['business_name', 'restaurant_type']].
    ↪drop_duplicates().sort_values(
        ['restaurant_type', 'business_name']))
display(restaurant_type_mapping)

```

	business_name	restaurant_type
297	Alsancak Firin	Bakery / Dessert
946	Kocak Baklava	Bakery / Dessert
407	Pecko Firin Atakum	Bakery / Dessert
649	Sevinc Pastanesi	Bakery / Dessert
1001	J Burber	Burger
...	...	...
935	Fikret Balik Restaurant	Restaurant
880	Inegol Restaurant	Restaurant
528	Kasikla Restaurant	Restaurant

803	Koco Restaurant	Restaurant
242	Tuana Restaurant	Restaurant

[100 rows x 2 columns]

11.2 Theme distribution by restaurant type

```
[59]: type_theme_counts = pd.crosstab(model_df['restaurant_type'],model_df['theme'])
type_theme_percent = (type_theme_counts.div( type_theme_counts.
↪sum(axis=1),axis=0) * 100)
display(type_theme_percent.round(1))
```

theme	Burger Experience	Cleanliness, Ambience & Environment \
restaurant_type		
Bakery / Dessert	0.0	0.0
Burger	48.5	3.0
Cafe	3.0	9.1
Dessert / Ice Cream	0.0	0.0
Kebab	4.5	4.5
Local Restaurant	0.0	7.3
Other / Unknown	5.5	6.4
Pizza	0.0	2.3
Restaurant	6.8	8.0

theme	Food Taste & Satisfaction \
restaurant_type	
Bakery / Dessert	22.7
Burger	0.0
Cafe	3.0
Dessert / Ice Cream	36.4
Kebab	13.6
Local Restaurant	1.8
Other / Unknown	7.8
Pizza	2.3
Restaurant	6.8

theme	Overall Dining Experience & Menu Variety \
restaurant_type	
Bakery / Dessert	13.6
Burger	9.1
Cafe	12.1
Dessert / Ice Cream	0.0
Kebab	9.1
Local Restaurant	21.8
Other / Unknown	16.1
Pizza	2.3
Restaurant	26.1

theme	Pizza Experience	Positive Overall Experience \
restaurant_type		
Bakery / Dessert	0.0	9.1
Burger	0.0	6.1
Cafe	12.1	21.2
Dessert / Ice Cream	0.0	9.1
Kebab	0.0	9.1
Local Restaurant	0.0	10.9
Other / Unknown	0.8	14.7
Pizza	72.7	9.1
Restaurant	0.0	15.9

theme	Price & Value	Regional / Specialty Food Experience \
restaurant_type		
Bakery / Dessert	22.7	13.6
Burger	12.1	6.1
Cafe	0.0	6.1
Dessert / Ice Cream	27.3	9.1
Kebab	36.4	13.6
Local Restaurant	14.5	21.8
Other / Unknown	11.4	11.0
Pizza	0.0	0.0
Restaurant	8.0	5.7

theme	Short Positive Praise / Recommendation \
restaurant_type	
Bakery / Dessert	2.3
Burger	3.0
Cafe	9.1
Dessert / Ice Cream	18.2
Kebab	0.0
Local Restaurant	9.1
Other / Unknown	7.0
Pizza	0.0
Restaurant	5.7

theme	Staff & Service Quality	Waiting Time & Service Friction
restaurant_type		
Bakery / Dessert	4.5	11.4
Burger	6.1	6.1
Cafe	6.1	18.2
Dessert / Ice Cream	0.0	0.0
Kebab	4.5	4.5
Local Restaurant	7.3	5.5
Other / Unknown	7.7	11.7
Pizza	2.3	9.1
Restaurant	8.0	9.1


```
[60]: type_operational_percent = (type_theme_percent[ operational_themes])
display(type_operational_percent.round(1))
```

theme	Price & Value	Waiting Time & Service Friction \
restaurant_type		
Bakery / Dessert	22.7	11.4
Burger	12.1	6.1
Cafe	0.0	18.2
Dessert / Ice Cream	27.3	0.0
Kebab	36.4	4.5
Local Restaurant	14.5	5.5
Other / Unknown	11.4	11.7
Pizza	0.0	9.1
Restaurant	8.0	9.1

theme	Staff & Service Quality \
restaurant_type	
Bakery / Dessert	4.5
Burger	6.1
Cafe	6.1
Dessert / Ice Cream	0.0
Kebab	4.5
Local Restaurant	7.3
Other / Unknown	7.7
Pizza	2.3
Restaurant	8.0

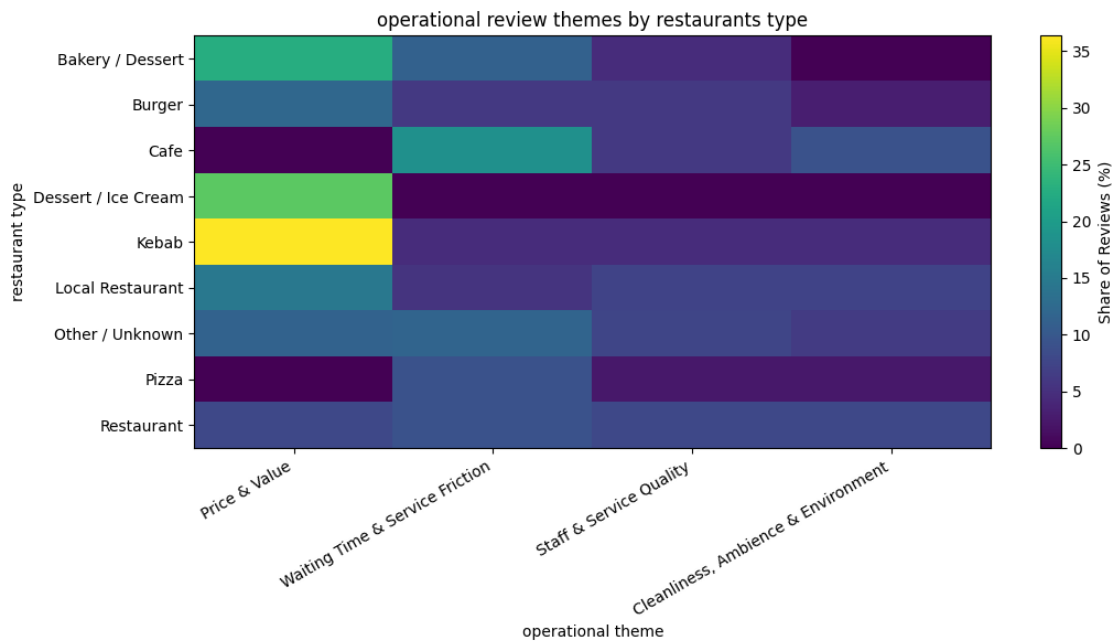
theme	Cleanliness, Ambience & Environment
restaurant_type	
Bakery / Dessert	0.0
Burger	3.0
Cafe	9.1
Dessert / Ice Cream	0.0
Kebab	4.5
Local Restaurant	7.3
Other / Unknown	6.4
Pizza	2.3
Restaurant	8.0

```
[61]: plt.figure(figsize=(11, 6))
plt.imshow(type_operational_percent,aspect='auto')
plt.colorbar(label='Share of Reviews (%)')

plt.xticks(range( len(type_operational_percent.
↪columns)),type_operational_percent.columns,rotation=30,ha='right')
```

```
plt.yticks(range(len(type_operational_percent.index)),type_operational_percent.
↪index)
plt.xlabel('operational theme')
plt.ylabel('restaurant type')
plt.title('operational review themes by restaurants type')

plt.tight_layout()
plt.show()
```



Make the comparison more trustworthy

```
[62]: type_support = (
    model_df.groupby('restaurant_type').agg(reviews=('text_original',
↪'count'),restaurants=('business_name', 'nunique'),average_rating=('rating',
↪'mean')) .sort_values('reviews',ascending=False).round(2))
display(type_support)
```

restaurant_type	reviews	restaurants	average_rating
Other / Unknown	770	70	3.91
Restaurant	88	8	3.78
Local Restaurant	55	5	4.09
Bakery / Dessert	44	4	3.82
Pizza	44	4	4.05
Cafe	33	3	4.12
Burger	33	3	3.73
Kebab	22	2	4.09

Dessert / Ice Cream	11	1	3.91
---------------------	----	---	------

1.14 12. Supported Restaurant Type Comparison&

The dataset does not directly provide the type of restaurant. Only when the company name included a clear indicator, like 'Pizza,' 'Cafe,' or 'Burger,' was a conservative type derived. Restaurant-type comparisons are regarded as exploratory rather than definitive since many restaurants cannot be categorized with confidence.

Only restaurant types represented by at least three restaurants are kept, and Other / Unknown is eliminated to minimize erratic comparisons.

```
[63]: type_restaurant_counts = (
        model_df[['business_name', 'restaurant_type']].
        ↪drop_duplicates()['restaurant_type'].value_counts()
supported_types = type_restaurant_counts[type_restaurant_counts >= 3].index.
        ↪tolist()
supported_types = [
    restaurant_type
    for restaurant_type in supported_types
    if restaurant_type != 'Other / Unknown']
print('Supported restaurant types:')
print(supported_types)
```

Supported restaurant types:

```
['Restaurant', 'Local Restaurant', 'Pizza', 'Bakery / Dessert', 'Cafe',
'Burger']
```

```
[64]: type_analysis_df = model_df[model_df['restaurant_type'].isin(supported_types)].
        ↪copy()
print('Reviews included:',len(type_analysis_df))
print('Restaurants included:',type_analysis_df['business_name'].nunique())
```

Reviews included: 297

Restaurants included: 27

12.2 Build the supported operational-theme table

```
[65]: supported_type_theme_counts = pd.
        ↪crosstab(type_analysis_df['restaurant_type'],type_analysis_df['theme'])
supported_type_theme_percent = (supported_type_theme_counts.
        ↪div(supported_type_theme_counts.sum(axis=1), axis=0)* 100)
supported_operational = (supported_type_theme_percent[operational_themes].
        ↪round(1))
display(supported_operational)
```

theme	Price & Value	Waiting Time & Service Friction	\
restaurant_type			
Bakery / Dessert	22.7		11.4
Burger	12.1		6.1

Cafe	0.0	18.2
Local Restaurant	14.5	5.5
Pizza	0.0	9.1
Restaurant	8.0	9.1

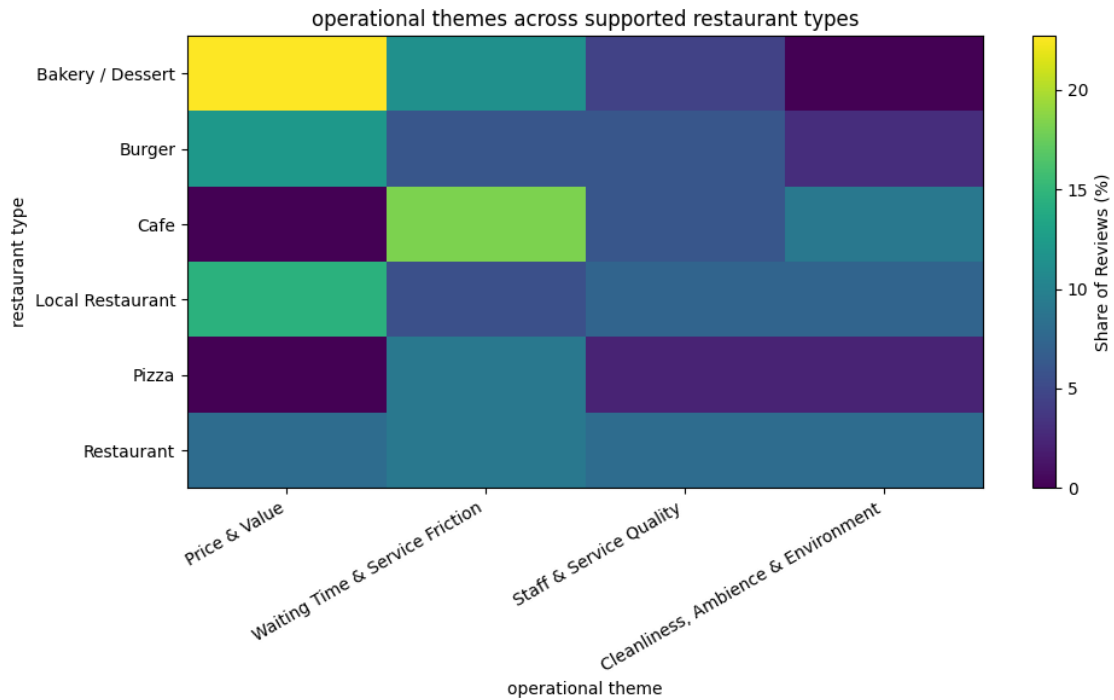
theme	Staff & Service Quality	Cleanliness, Ambience & Environment
restaurant_type		
Bakery / Dessert	4.5	0.0
Burger	6.1	3.0
Cafe	6.1	9.1
Local Restaurant	7.3	7.3
Pizza	2.3	2.3
Restaurant	8.0	8.0

```
[66]: supported_type_summary = (type_analysis_df.groupby('restaurant_type').
    ↪agg(restaurants=('business_name', 'nunique'),reviews=('text_original',
    ↪'count'),average_rating=('rating', 'mean')).round(2).
    ↪sort_values('restaurants',ascending=False))
display(supported_type_summary)
```

	restaurants	reviews	average_rating
restaurant_type			
Restaurant	8	88	3.78
Local Restaurant	5	55	4.09
Pizza	4	44	4.05
Bakery / Dessert	4	44	3.82
Burger	3	33	3.73
Cafe	3	33	4.12

12.3 Make a better heatmap

```
[67]: plt.figure(figsize=(10, 6))
plt.imshow(supported_operational, aspect='auto')
plt.colorbar(label='Share of Reviews (%)')
plt.xticks(range(len(supported_operational.columns)),supported_operational.
    ↪columns,rotation=30,ha='right')
plt.yticks( range(len(supported_operational.index)),supported_operational.index)
plt.xlabel('operational theme')
plt.ylabel('restaurant type')
plt.title('operational themes across supported restaurant types')
plt.tight_layout()
plt.show()
```



12.4 Calculate how unusual each type is

```
[68]: platform_theme_percent = ( model_df['theme'].value_counts(normalize=True).
    ↳mul(100))
```

```
platform_operational = (platform_theme_percent[operational_themes])
display(platform_operational.round(1).to_frame('platform_percent'))
```

```
platform_percent
theme
Price & Value                11.6
Waiting Time & Service Friction 10.8
Staff & Service Quality        7.1
Cleanliness, Ambience & Environment 6.0
```

```
[69]: type_theme_lift = (supported_operational.subtract(
    ↳platform_operational,axis='columns'))
display( type_theme_lift.round(1))
```

```
theme      Price & Value  Waiting Time & Service Friction \
restaurant_type
Bakery / Dessert      11.1                0.6
Burger                0.5               -4.7
Cafe                 -11.6                7.4
Local Restaurant       2.9               -5.3
Pizza                -11.6               -1.7
```

Restaurant	-3.6	-1.7
theme	Staff & Service Quality	Cleanliness, Ambience & Environment
restaurant_type		
Bakery / Dessert	-2.6	-6.0
Burger	-1.0	-3.0
Cafe	-1.0	3.1
Local Restaurant	0.2	1.3
Pizza	-4.8	-3.7
Restaurant	0.9	2.0

12.5 Visualize the lift

```
[70]: plt.figure(figsize=(10, 6))

plt.imshow( type_theme_lift,aspect='auto')
plt.colorbar(label='Difference from Platform Average (percentage points)')
plt.xticks(range(len(type_theme_lift.columns)), type_theme_lift.columns,
           rotation=30,ha='right')

plt.yticks( range(len(type_theme_lift.index)),type_theme_lift.index)
plt.xlabel('operational theme')
plt.ylabel('restaurant type')
plt.title('which operational themes over or Under Index by Restaurant Type?')

plt.tight_layout()
plt.show()
```



12.6 Automatically extract the strongest deviations

```
[71]: type_lift_long = (type_theme_lift.reset_index().melt(
    ↪id_vars='restaurant_type',var_name='theme',value_name='lift_percentage_points'))

type_lift_long[ 'absolute_lift'
] = ( type_lift_long['lift_percentage_points'].abs())

strongest_type_patterns = (type_lift_long.
    ↪sort_values('absolute_lift',ascending=False).head(10))

display(strongest_type_patterns[ [ 'restaurant_type',
    ↪'theme','lift_percentage_points']].round(1))
```

	restaurant_type	theme \
2	Cafe	Price & Value
4	Pizza	Price & Value
0	Bakery / Dessert	Price & Value
8	Cafe	Waiting Time & Service Friction
18	Bakery / Dessert	Cleanliness, Ambience & Environment
9	Local Restaurant	Waiting Time & Service Friction
16	Pizza	Staff & Service Quality
7	Burger	Waiting Time & Service Friction
22	Pizza	Cleanliness, Ambience & Environment
5	Restaurant	Price & Value

	lift_percentage_points
2	-11.6
4	-11.6
0	11.1
8	7.4
18	-6.0
9	-5.3
16	-4.8
7	-4.7
22	-3.7
5	-3.6

1.15 13. Stakeholder Findings

Price, waiting time and service friction, service quality, and cleanliness/ambience are some of the operational themes that are common to all restaurants, according to the final analysis. Because only 27 restaurants could be categorized into supported types with at least three restaurants per category, restaurant type comparisons were treated as exploratory. ### Finding 1 - Waiting-time issues are widespread, but especially elevated in cafés

About 10.8% of reviews on the platform were related to waiting time and service friction. Cafés had the highest rate of any supported restaurant type (18.2%), 7.4 percentage points higher than the platform baseline. This implies that for cafés in the sample that is currently available, waiting times, responsiveness, or service flow might be particularly significant operational concerns.

1.15.1 Finding 2 - Bakery and dessert businesses show stronger price/value concerns

Approximately 11.6% of reviews on the entire platform were related to price and value. The rate for bakeries and dessert shops was 22.7%, which is about 11.1 percentage points higher than the platform baseline. Customers may be especially sensitive to perceived value in bakery and dessert contexts, as indicated by the largest positive deviation found among the supported restaurant types.

Finding 3 - review experience themes can also concentrate at individual restaurants

Certain themes were highly concentrated at particular businesses, according to the restaurant level analysis. For instance, Nesh received seven out of eleven reviews related to Waiting Time & Service Friction, whereas Hacı'nin Yeri Yigit Lokantasi received six out of eleven reviews related to Price & Value. These findings imply that certain operational issues might be unique to restaurants rather than the industry as a whole.

1.15.2 Finding 4 - Semantic clustering produced both operational and cuisine-specific structure

Both broad operational themes and cuisine-specific groups like Pizza Experience and Burger Experience were found by the final clustering model. This suggests that the review text has a deeper structure than just positive and negative sentiments. The model was able to distinguish between the type of food experience that patrons were talking about and how they felt about the restaurant.

1.15.3 Finding 5 - Findings should be used as outreach signals, not definitive restaurant rankings

Since there are only 11 reviews for each restaurant, one review makes up roughly 9.1% of the sample that is available. Therefore, based on just one or two reviews, restaurant level percentages can vary significantly. Rather than being definitive indicators of restaurant performance, the results are best understood as signals that can assist the outreach team in determining where additional research or discussion may be helpful.

1.16 14. Assumptions Log

The analysis relies on several assumptions that affect how the results should be interpreted:-

Assumption	Reason	Potential Impact
All 1,100 reviews are included in theme discovery	The client requested analysis of all reviews rather than only negative ones	Positive and negative experiences may appear within the same semantic theme
Ratings are used only for interpretation, not clustering	The goal is to discover themes naturally from review text	Clusters may contain mixed sentiment
Repeated short reviews such as 'Great' or 'Expensive' are retained	They occur across different restaurants and may represent genuine independent reviews	Very short expressions may influence some clusters

Assumption	Reason	Potential Impact
Restaurant-level percentages are based on 11 reviews per restaurant	The dataset contains exactly 11 reviews for each restaurant	One review changes a restaurant-level percentage by about 9.1 percentage points
Restaurant type is derived conservatively from business names	The dataset does not provide verified restaurant-type metadata	Restaurant-type findings are exploratory and may not cover all businesses accurately
Only restaurant types represented by at least three restaurants are compared	Very small groups would produce unstable percentages	Some potentially interesting types are excluded
Cluster labels are assigned through human interpretation	Unsupervised models do not generate stakeholder-ready labels automatically	Theme naming involves judgment
Semantic similarity is treated as evidence of thematic similarity	Sentence embeddings encode overall meaning	Reviews discussing multiple issues may still be assigned to one dominant cluster

These assumptions were documented before drawing stakeholder conclusions so that model outputs are not interpreted as stronger evidence than the dataset supports.

1.17 15. Limitations

1.17.1 Small restaurant-level samples

Restaurant level percentages can vary significantly based on just one or two reviews because each restaurant has only 11 reviews. Therefore, rather than being exact indicators of restaurant performance, the concentrations found should be regarded as early warning signs.

1.17.2 Restaurant-type metadata

There isn't a verified restaurant-type field in the dataset. Many restaurants were left unclassified because restaurant types were conservatively derived from obvious business name indicators. Therefore, restaurant type comparisons should be considered exploratory as they only cover a portion of the dataset.

1.17.3 Multi-topic reviews

The final K-Means model assigns each review to a single cluster, so secondary themes within the same review may not be represented. However, a single review can address multiple issues at once, such as food quality, waiting time, price, and staff behaviour.

1.17.4 Human interpretation of clusters

Groups of reviews that are semantically similar are found by the clustering model, but their business meaning is not automatically ascertained. After examining representative reviews, keywords, rating trends, and restaurant coverage, final theme names were chosen.

1.17.5 Cluster quality

Silhouette scores were still comparatively low, but the semantic K-Means model outperformed the TF-IDF baseline. This shows theme overlap, which is to be expected given that dining out frequently involves related aspects like food, service, cost, and ambiance.

1.17.6 Density-based clustering trade-off

Although UMAP + HDBSCAN generated many small clusters and identified a significant percentage of reviews as noise, it also produced some highly specific complaint groups and stronger numerical separation. Because of this, it was not used as the final model but instead kept as a supporting experiment.

1.17.7 Dataset representativeness

The dataset may not accurately reflect Turkey’s entire restaurant industry because it only includes reviews for a small sample of eateries. Without more proof, conclusions shouldn’t be extrapolated from the data at hand.

1.18 16. Executive Summary

This project examined 1,100 restaurant reviews from 100 restaurants in order to find recurrent themes related to the customer experience and ascertain whether certain problems are common to all restaurants or are specific to certain types of businesses and restaurants. SentenceTransformer embeddings were used to compare semantic clustering with a baseline of TF-IDF + K-Means. SentenceTransformer embeddings with K-Means were chosen as the final model because the semantic approach generated more cohesive and useful themes.

The final analysis identified 11 themes, including four especially relevant operational themes: **Price & Value**, **Waiting Time & Service Friction**, **Staff & Service Quality**, and **Cleanliness, Ambience & Environment** because they were the most directly actionable for restaurant outreach.

There were a few distinct patterns. In the exploratory restaurant-type comparison, Price & Value appeared in 67% of restaurants, while Waiting Time & Service Friction appeared in 66% of restaurants and was strongly concentrated at some individual businesses. While bakery and dessert businesses showed significantly more price/value discussion, cafes showed more waiting/service-friction discussion than the platform average.

The findings imply that there is an uneven distribution of customer complaints among different types of restaurants. As a result, the outreach team can use the themes as indicators to pinpoint areas where particular operational problems might need more research or focused messaging. These results should be regarded as exploratory decision support signals rather than final assessments of restaurant performance because each restaurant only has 11 reviews and restaurant-type metadata is lacking.

1.19 17. Conclusion

The analysis shows that unsupervised semantic clustering can convert unprocessed restaurant reviews into comprehensible themes that are more valuable than just ratings. Price/value, waiting time, staff/service quality, cleanliness or environment were the strongest operational patterns.

While some themes were heavily concentrated at specific establishments, others were prevalent throughout restaurants. Additionally, the project demonstrated the significance of model comparison. Semantic embeddings enhanced thematic coherence, TF-IDF offered a transparent baseline, and HDBSCAN identified niche patterns but added significant fragmentation and noise.

All things considered, the final semantic K-Means model offered the best compromise between stakeholder utility, interpretability, and review coverage.

```
[72]: streamlit_export = model_df[ ["business_name","text_original","rating",
    ↪ "rating_category","rating_group", "review_word_count", "final_cluster",
    ↪ "theme", "theme_confidence","restaurant_type",]
].copy()
streamlit_export.to_csv("serafino_reviews.csv",index=False,)
print("Saved:", streamlit_export.shape)
display(streamlit_export.head())
```

Saved: (1100, 10)

```

                business_name \
0  Haci'nin Yeri - Yigit Lokantasi
1  Haci'nin Yeri - Yigit Lokantasi
2  Haci'nin Yeri - Yigit Lokantasi
3  Haci'nin Yeri - Yigit Lokantasi
4  Haci'nin Yeri - Yigit Lokantasi
```

```

                text_original \
0  We went to Marmaris with my wife for a holiday. We chose this restaurant as a
    ↪ place for dinner based on the reviews and because we wanted juicy fo...
1  During my holiday in Marmaris we ate here to fit the food. It's really good
    ↪ that the food is cheap and nice. Eating as much bread as you want is a...
2  Prices are very affordable. The menu in the photo cost 108 liras. You have to
    ↪ wait 10-15 minutes for food. Staff is annoying. Well it tastes good...
3
    ↪ Turkey's cheapest artisan restaurant and its food is delicious!
4  I don't know what you will look for in terms of price performance point;
    ↪ taste; but yigit restaurant writes a big plus for those who come to work ...
```

```

rating    rating_category    rating_group    review_word_count \
0         5              taste  POSITIVE (4-5)             130
1         4              menu   POSITIVE (4-5)             179
2         3  outdoor_atmosphere    MIXED (3)              31
3         5  indoor_atmosphere  POSITIVE (4-5)              9
4         3              menu   MIXED (3)              31
```

```

final_cluster    theme    theme_confidence \
0              2  Overall Dining Experience & Menu Variety    Medium
1              3              Price & Value                High
2              3              Price & Value                High
```

3	1	Regional / Specialty Food Experience	Medium
4	3	Price & Value	High

```

restaurant_type
0 Local Restaurant
1 Local Restaurant
2 Local Restaurant
3 Local Restaurant
4 Local Restaurant

```

Links-

Presentation - https://drive.google.com/drive/u/0/folders/1crF_Tod-MzpZwBCCUmAk3YGkgJXltjlR

StreamLit- <https://data-science-for-human-centered-systems-serafinosignal.streamlit.app/>